Supplementary Material for: A Data-Driven, Component-Level Projection Framework for 21st-Century Sea-Level Rise

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1 Materials and Methods

Unless otherwise noted, all reported intervals are 90% Bayesian credible intervals (equal-tailed), and where 1σ is reported, we state explicitly that it is a standard deviation.

1.1 Observational datasets

All component models are calibrated against published observational products (Table S1). We use:

- **Global mean sea level:** NASA satellite altimetry (TOPEX/Jason, 1993–2025; *Beckley et al.*, 2017; *Hamlington et al.*, 2020), with the Frederikse et al. (2020) tide-gauge reconstruction (1900–2018) and the Dangendorf et al. (2024) reconstruction (1900–2021) for pre-satellite validation. None of these records enters any component calibration.

- **Ocean thermal expansion:** NOAA thermosteric sea level (doi:10.7289/V53F4MVP), 0–700 m (1955–2025) and 0–2000 m (2005–2025) (*Levitus et al.*, 2012). Both records are scaled by 1.14 before fitting. This corrects the 14% low bias that *Li et al.* (2022) find in thermosteric trends computed with linear vertical interpolation. *Li et al.* (2022) quantify this bias for the upper 700 m; applying the same factor to the 0–2000 m record is our assumption. A residual structural uncertainty of 5% of the corrected signal is retained. The fit uses the 0–700 m record and a 700–2000 m record, formed as the difference between the 0–2000 m and 0–700 m records in the same years, so that the 0–700 m data enter the fit once. The volcanic term of the upper layer uses the annual volcanic effective radiative forcing of *Forster et al.* (2025) (data release v2025.06.11, doi:10.5281/zenodo.15639576), held at zero after 2024. The 700–2000 m layer is carried by the 0–2000 m record from 2005 onward and by the deep layer of the two-layer model at earlier times. A constant 0.12 ± 0.04 mm yr^{-1} (one standard deviation, 1991–2010) (*Desbruyères et al.*, 2016) is added over the full record for the contribution below 2000 m. Two reconstructions of global-mean steric sea level are withheld for validation, from Frederikse et al. (2020) (1900–2018) and Dangendorf et al. (2024) (1900–2021). Over 1957–2018 the Frederikse et al. reconstruction draws on the same archive of in situ profiles as the NOAA record (*Levitus et al.*, 2012; *Ishii et al.*, 2017; *Cheng and Zhu*, 2016), so it is not fully independent of it there. The Dangendorf et al. reconstruction uses no in situ ocean temperature or salinity. Its steric term is the residual of a sea-level budget whose barystatic input is from *Frederikse et al.* (2020). Neither reconstruction carries the 1.14 scaling. Thermosteric sea level computed from the EN4.2.2 analyses (*Good et al.*, 2013) and the IAP v4.2 ocean heat content (*Cheng et al.*, 2017) replace the NOAA records only in the sensitivity test of Section 1.9.

- **Glaciers:** GlaMBIE global consensus (The GlaMBIE Team 2025), combining glaciological field measurements, satellite altimetry, DEM differencing, and GRACE/GRACE-FO gravimetry across all 19 RGI regions. We use 2000–2023 (24 annual points) for calibration. For validation we use the global glacier mass changes of *Zemp et al.* (2019) for 1962–1999 (38 hydrological years).

- **Greenland:** *Mouginot et al.* (2019) SMB and discharge (1972–2018), of which the 41 years from 1978 enter the discharge fit, the years covered by the delayed ocean forcing at every candidate lag. *Mankoff et al.* (2021) discharge (1986–2023) and the IMBIE-3 Greenland record (1972–2023; *Otosaka et al.* (2026)) are withheld for validation. Subsurface ocean temperature from EN4 v4.2.2 with g10 bias corrections (200–500 m, Greenland peripheral waters, 1970–2021; *Good et al.* (2013)). EN4 over 1958–1969 is withheld for validation of the surface-to-ocean transfer function. Surface temperature from Berkeley Earth (*Rohde and Hausfather*, 2020). Greenland SMB is projected with a statistical emulator of the regional climate model MARv3.12 driven by seven CMIP6 global climate models (CESM2, MPI-ESM1-2-HR, NorESM2-MM, UKESM1-0-LL, CNRM-CM6-1, CNRM-ESM2-1, and IPSL-CM6A-LR). These simulations were produced for the PROTECT project (*Boberg et al.*, 2026) and are publicly available at `http://ftp.climato.be/fettweis/MARv3.12/Greenland/PROTECT`. We use the annual SMB of the contiguous ice sheet, excluding peripheral glaciers, over 1980–2014 and over each available SSP to 2100, together with the global mean surface temperature of each driving model from the CMIP6 archive. *Mankoff et al.* (2021) SMB (1986–2023), the mean of three regional climate models driven by reanalysis, is withheld for validation of the emulator.

- **Antarctica:** IMBIE-3 mass balance (1979–2023, 45 annual points per region; *Otosaka et al.* (2026)), partitioned into West Antarctica, East Antarctica, and the Antarctic Peninsula.

- **Terrestrial water storage:** *Frederikse et al.* (2020) reconstruction, 1900–2018, used with its reported uncertainty in the budget closure and the hindcast. From 2003 it derives from GRACE and GRACE-FO observations. Before 2003 it is the sum of water impoundment in artificial reservoirs, groundwater depletion, and natural variability in storage, estimated without gravimetric observations (Fig. S10).

- **Temperature scenarios:** IPCC AR6 assessed GMST projections (SSP1-1.9, SSP1-2.6, SSP2-4.5, SSP3-7.0, SSP5-8.5; *Lee et al.* (2021)).

Table S1: Datasets used in the component framework and the role each plays. Records marked *validation* are withheld from every component calibration, including observed GMSL itself. A complete dataset inventory including versions, spatial domains, and processing details is provided in `data/dataset_inventory.xlsx`.

Component	Dataset	Span	Role	Reference
GMSL	NASA altimetry	1993–2025	Validation	Beckley et al. (2017)
GMSL	Frederikse reconstruction	1900–2018	Validation	Frederikse et al. (2020)
GMSL	Dangendorf reconstruction	1900–2021	Validation	Dangendorf et al. (2024)
Thermosteric	NOAA thermosteric, 0–700 m	1955–2025	Calibration	Levitus et al. (2012)
Thermosteric	NOAA thermosteric, 0–2000 m	2005–2025	Calibration, as 700–2000 m	Levitus et al. (2012)
Thermosteric	Volcanic effective radiative forcing	1850–2024	Forcing	Forster et al. (2025)
Thermosteric	Frederikse steric	1900–2018	Validation	Frederikse et al. (2020)
Thermosteric	Dangendorf sterodynamic	1900–2021	Validation	Dangendorf et al. (2024)
Thermosteric	EN4.2.2 thermosteric	1955–2024	Sensitivity test	Good et al. (2013)
Thermosteric	IAP v4.2 heat content	1955–2024	Sensitivity test	Cheng et al. (2017)
Glaciers	GlaMBIE global consensus	2000–2023	Calibration	The GlaMBIE Team (2025)
Glaciers	Zemp glaciological and geodetic	1962–1999	Validation	Zemp et al. (2019)
Greenland	Mouginot discharge & SMB	1972–2018	Calibration	Mouginot et al. (2019)
Greenland	EN4 subsurface ocean T	1970–2021	Forcing	Good et al. (2013)
Greenland	EN4 subsurface ocean T	1958–1969	Validation	Good et al. (2013)
Greenland	Mankoff PROMICE discharge & SMB	1986–2023	Validation	Mankoff et al. (2021)
Greenland	IMBIE-3 Greenland	1972–2023	Validation	Otosaka et al. (2026)
Greenland	MARv3.12 SMB, seven CMIP6 GCMs	1980–2100	Emulator training	Boberg et al. (2026)
Greenland	GMST of the seven driving GCMs	1950–2100	Emulator training	CMIP6 archive
Peninsula	IMBIE-3 Peninsula	1979–2023	Calibration	Otosaka et al. (2026)
EAIS	IMBIE-3 East Antarctica	1979–2023	Diagnostic	Otosaka et al. (2026)
WAIS	IMBIE-3 West Antarctica	1979–2023	Calibration	Otosaka et al. (2026)
TWS	Frederikse TWS	1900–2018	Forcing	Frederikse et al. (2020)
Temperature	Berkeley Earth surface T	1850–2024	Forcing	Rohde & Hausfather (2020)
Temperature	IPCC AR6 GMST projections	2015–2099	Forcing	Lee et al. (2021)
Projections	IPCC AR6 FACTS	2020–2150	Comparison	Garner et al. (2023); Kopp et al. (2023)

1.2 Variables and parameters

Table S2 lists every symbol that appears in the main text, with a plain-language description, the value where one is assigned, units, and the source of any assigned value. Symbols used only in this supplement are defined where they appear there and are not repeated here.

Table S2: **Symbols used in the main text.** The value column gives the value of parameters that are assigned rather than inferred. Quantities we calibrate are marked *calibrated* and their posterior medians and 90% credible intervals are reported in the main text, since a fitted value is not a prescribed one. Model variables that carry no single value are marked with a dash. Where one symbol carries more than one meaning, each meaning is listed separately and its context named. SLE denotes sea-level equivalent.

Symbol	Description	Value	Units	Source
General				
t	Time	—	yr	—
T	Global mean surface temperature (GMST) anomaly, relative to 1995–2005 in every model equation (scenario warming levels are relative to 1850–1900)	—	°C	*Rohde and Hausfather* (2020)
t_0	Study baseline year, to which cumulative contributions are referenced in the component models	2000	yr	—
t_0	Forecast origin, the most recent year of satellite altimetry in the blended forecast	2025	yr	—
i	Index over observations, outlet glaciers, or Monte Carlo samples	—	—	—
z	Depth below the sea surface	—	m	—
Greenland Ice Sheet discharge				
D	Greenland solid-ice discharge, as a contribution to sea level	—	$\mathrm{mm\,SLE\,yr^{-1}}$	—
H_{D}	Cumulative Greenland discharge contribution to GMSL	—	mm SLE	—
γ	Sensitivity of discharge to subsurface ocean temperature	calibrated	$\mathrm{mm\,SLE\,yr^{-1}\,°C^{-1}}$	this study
r_0	Background discharge rate at the baseline ocean temperature	calibrated	$\mathrm{mm\,SLE\,yr^{-1}}$	this study
δ	Lag between ocean warming and the discharge response	6, selected by BIC	yr	this study
T_{ocean}	Subsurface ocean temperature anomaly, 200–500 m, in Greenland's peripheral waters	—	°C	*Good et al.* (2013)
Dimensional scaling for γ				
$\widetilde{\gamma}$	Order-of-magnitude estimate of γ from the scaling argument	0.05–1.5	$\mathrm{mm\,SLE\,yr^{-1}\,°C^{-1}}$	this study
$\Delta\dot{m}_i$	Submarine melt rate of outlet glacier i	—	$\mathrm{m\,yr^{-1}}$	—

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Symbol	Description	Value	Units	Source
κ_i	Melt sensitivity to ocean thermal forcing, averaged over the calving face	36–104, geometric mean 61	$\mathrm{m\,yr^{-1}\,{}^{\circ}C^{-1}}$	*Xu et al.* (2013); *Rignot et al.* (2016); *Slater and Straneo* (2022); *Wood et al.* (2021)
μ_i	Ratio of the total ice-flux perturbation to the direct submarine melt flux	3–10	—	*Enderlin and Howat* (2013); *Wood et al.* (2021)
$A_{\mathrm{f},i}$	Submerged calving-front area of outlet glacier i	0.9–2.7	$\mathrm{km^2}$	*Rignot et al.* (2016)
$\Delta\dot{Q}_i$	Ice-flux perturbation from outlet glacier i	—	$\mathrm{Gt\,yr^{-1}}$	—
N	Number of marine-terminating outlet glaciers resolved in the record	219	—	*Mouginot et al.* (2019)
ρ_{ice}	Density of ice	900	$\mathrm{kg\,m^{-3}}$	—
ρ_{fw}	Density of freshwater	1000	$\mathrm{kg\,m^{-3}}$	—
A_{ocean}	Global ocean area, setting the convention $1\,\mathrm{mm\,SLE} = 362.5\,\mathrm{Gt}$	3.625×10^{14}	$\mathrm{m^2}$	—
Thermosteric two-layer model				
H_{u}	Steric height of the upper ocean layer	—	mm	—
H_{ud}	Steric height of the upper 2000 m	—	mm	—
H_{full}	Full-depth thermosteric contribution to GMSL	—	mm	—
α_{sw}	Thermal expansion coefficient of seawater	—	$\mathrm{{}^{\circ}C^{-1}}$	—
α'_{sw}	Derivative of α_{sw} with respect to temperature	—	$\mathrm{{}^{\circ}C^{-2}}$	—
$\Delta\theta$	Ocean temperature anomaly profile	—	°C	—
φ	Dimensionless shape factor of the temperature anomaly profile	—	—	—
h_{u}	Thickness of the upper layer	700	m	—
h_{d}	Lower bound of the deep layer	2000	m	—
S_{u}	Thermal state of the upper layer, the amplitude of its temperature anomaly	—	°C	—
S_{d}	Thermal state of the deep layer	—	°C	—
a_{u}	Quadratic expansion coefficient of the upper layer, $\kappa_{\mathrm{sw}} b_{\mathrm{u}}^2$	derived	$\mathrm{mm\,{}^{\circ}C^{-2}}$	this study
κ_{sw}	Equation-of-state coefficient tying a_{u} to b_{u}, from TEOS-10 and EN4	0.265	$\mathrm{m^{-1}}$	*IOC, SCOR and IAPSO* (2010); *Good et al.* (2013)
g	Shape of the upper-layer anomaly profile, $\varphi = A\,g$, uniform over 0–700 m	—	—	—

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Symbol	Description	Value	Units	Source
b_{u}	Linear expansion coefficient of the upper layer	calibrated	$\mathrm{mm\,°C^{-1}}$	this study
b_{d}	Linear expansion coefficient of the deep layer	calibrated	$\mathrm{mm\,°C^{-1}}$	this study
τ_{u}	Relaxation time of the upper layer toward GMST	calibrated	yr	this study
τ_{d}	Relaxation time of the deep layer toward the upper layer	calibrated	yr	this study
F_{v}	Volcanic effective radiative forcing, zero after 2024	—	$\mathrm{W\,m^{-2}}$	*Forster et al.* (2025)
k_{v}	Coefficient of the volcanic term, $m_{\mathrm{v}}c_{\mathrm{v}}/b_{\mathrm{u}}$	derived	$\mathrm{°C\,yr^{-1}}$ per $\mathrm{W\,m^{-2}}$	this study
c_{v}	0–700 m thermosteric rise per $\mathrm{W\,m^{-2}}$ of forcing over the ocean for one year	1.57	mm per $\mathrm{W\,m^{-2}\,yr}$	*Levitus et al.* (2012)
m_{v}	Multiplier of the volcanic term	calibrated	—	this study
$\dot{H}_{\mathrm{b}}$	Rate of steric expansion below 2000 m, adopted rather than calibrated	0.12 ± 0.04	$\mathrm{mm\,yr^{-1}}$	*Desbruyères et al.* (2016)
t_{ref}	Reference year for the below-2000 m term	2000	yr	—
$\sigma_{\mathrm{extra,u}}$, $\sigma_{\mathrm{extra,d}}$	Noise terms absorbing misfit beyond the reported observational error, 0–700 m and 700–2000 m	calibrated	mm	this study
ρ	First-order autoregressive coefficient of the noise terms, shared by both records	calibrated	—	this study
$\boldsymbol{\phi}$	Vector of the eight sampled thermosteric parameters	—	—	—
$\mathcal{J}$	Objective function, twice the negative log-posterior up to a constant	—	—	—
$\mathbf{r}_j$	Residuals between the demeaned model and record j (j = u: 0–700 m; j = d: 700–2000 m)	—	mm	—
$\sigma_{j,i}$	Observational error of point i of record j, formal and structural	—	mm	—
$\mathbf{C}_j$, $\mathbf{R}_j$	Residual covariance of record j and the autoregressive correlation of its noise term	—	$\mathrm{mm^2}$, —	—
$\mu_{\tau_{\mathrm{u}}}$	Mean of the log-normal prior on $\ln \tau_{\mathrm{u}}$	$\ln 8$	ln(yr)	*Geoffroy et al.* (2013)
$s_{\tau_{\mathrm{u}}}$	Standard deviation of the log-normal prior on $\ln \tau_{\mathrm{u}}$	0.5	—	*Geoffroy et al.* (2013)
$\mu_{\tau_{\mathrm{d}}}$	Mean of the log-normal prior on $\ln \tau_{\mathrm{d}}$, from the 16 CMIP5 models	$\ln 133$	ln(yr)	*Geoffroy et al.* (2013)

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Symbol	Description	Value	Units	Source
s_{τ_d}	Standard deviation of the log-normal prior on $\ln \tau_d$	0.47	—	*Geoffroy et al.* (2013)
s_m	Standard deviation of the normal prior on m_v, mean 1	0.19	—	*Forster et al.* (2021); *Fox-Kemper et al.* (2021)
Glaciers				
H_{gl}	Cumulative glacier contribution to GMSL	—	mm	—
b_{gl}	Sensitivity of the glacier loss rate to warming	calibrated	$mm\,yr^{-1}\,°C^{-1}$	this study, prior from *Rounce et al.* (2023)
c_{gl}	Glacier loss rate at the 1995–2005 mean GMST	$-b_{gl}T_{gl,eq}$	$mm\,yr^{-1}$	this study
$T_{gl,eq}$	GMST at which the glacier loss rate is zero (1850 balance), relative to 1995–2005	calibrated, prior $\mathcal{N}(-0.86, 0.15^2)$	°C	*Roe et al.* (2021), this study
$H_{gl,0}$	Glacier level at the baseline year	calibrated	mm	this study
ε_{gl}	Residual of the glacier fit	—	mm	—
$\sigma_{gl,extra}$	Model-inadequacy term added in quadrature to the annual GlaMBIE uncertainty	calibrated	$mm\,yr^{-1}$	this study
Σ_{gl}	Covariance matrix of the cumulative glacier record	—	mm^2	*The GlaMBIE Team et al.* (2025)
Antarctic Peninsula				
$\dot{H}_p$	Rate of the Antarctic Peninsula contribution	—	$mm\,yr^{-1}$	—
a_p	Quadratic sensitivity to warming, set to zero (Section 1.7)	0	$mm\,yr^{-1}\,°C^{-2}$	this study
b_p	Sensitivity of the Peninsula rate to warming	calibrated	$mm\,yr^{-1}\,°C^{-1}$	this study
c_p	Peninsula rate at the 1995–2005 mean GMST	calibrated	$mm\,yr^{-1}$	this study
$H_{p,0}$	Peninsula level at the first year of the record (1979)	calibrated	mm	this study
$\sigma_{p,extra}$	Model-inadequacy term for annual scatter beyond the reported uncertainty	calibrated	$mm\,yr^{-1}$	this study
East Antarctic Ice Sheet				
b_e	Sensitivity of the EAIS rate to warming, fitted as a check and not used in the forecast	calibrated	$mm\,yr^{-1}\,°C^{-1}$	this study, prior from *Frieler et al.* (2015); *Ligtenberg et al.* (2013)
c_e	EAIS rate at the 1995–2005 mean GMST	calibrated	$mm\,yr^{-1}$	this study
$H_{e,0}$	EAIS level at the first year of the record	calibrated	mm	this study

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Symbol	Description	Value	Units	Source
West Antarctic Ice Sheet				
a_{w}	Acceleration of WAIS mass loss in the quadratic-in-time fit	calibrated	$\mathrm{mm\,yr^{-2}}$	this study
$p(\mathrm{S1})$	Probability assigned to WAIS Scenario 1, the control variable of the scenario mixture	0.90 slow, 0.10 fast	—	this study
λ	Shape parameter of the skew-normal S2 distribution in log space	3	—	this study, motivated by *Robel et al.* (2019)
β	Exponent of the power law in time followed by the S2 trajectory	median 2, lognormal with log-sd 0.3	—	this study
Surface mass balance				
$\dot{H}_{\mathrm{smb}}$	Surface mass balance rate. For Greenland it is expressed as a contribution to sea-level rise	—	$\mathrm{Gt\,yr^{-1}}$ (EAIS), $\mathrm{mm\,SLE\,yr^{-1}}$ (Greenland)	—
a_{gris}	Quadratic sensitivity of the Greenland SMB contribution to warming	fitted to MAR for each of seven GCMs (Fig. S6)	$\mathrm{mm\,SLE\,yr^{-1}\,{}^{\circ}C^{-2}}$	this study, from MARv3.12
b_{gris}	Linear sensitivity of the Greenland SMB contribution to warming	fitted to MAR for each of seven GCMs (Fig. S6)	$\mathrm{mm\,SLE\,yr^{-1}\,{}^{\circ}C^{-1}}$	this study, from MARv3.12
c_{gris}	Greenland SMB contribution at zero GMST anomaly, fitted with the drift state	-1.07 [-1.41, -0.75]	$\mathrm{mm\,SLE\,yr^{-1}}$	this study, from *Mouginot et al.* (2019)
s	Drift state, the slowly varying departure of observed SMB from the emulator	fitted	$\mathrm{Gt\,yr^{-1}}$	this study, from *Mouginot et al.* (2019)
τ_{s}	Relaxation timescale of the drift state, set by a lognormal prior with 90% range 5–25 yr	12 [6, 26]	yr	this study
σ_s	Standard deviation of the drift state	108 [60, 186]	$\mathrm{Gt\,yr^{-1}}$	this study
σ_ε	Standard deviation of year-to-year weather in Greenland SMB	50 [10, 80]	$\mathrm{Gt\,yr^{-1}}$	this study
ϵ_{fb}	Fractional SMB–elevation feedback at the reference simulation, with negative draws set to zero	0.08 ± 0.05	—	*Fettweis et al.* (2013)
C_T	Linear sensitivity of EAIS SMB to GMST, positive for mass gain by accumulation	60 ± 20	$\mathrm{Gt\,yr^{-1}\,{}^{\circ}C^{-1}}$	*Frieler et al.* (2015); *Ligtenberg et al.* (2013)
C_{T^2}	Quadratic sensitivity of EAIS SMB, set to zero	0	$\mathrm{Gt\,yr^{-1}\,{}^{\circ}C^{-2}}$	this study
$\dot{M}_0$	Baseline EAIS SMB, set to zero because IMBIE-3 shows EAIS near balance	0	$\mathrm{Gt\,yr^{-1}}$	*Otosaka et al.* (2026)
Greenland temperature chain				

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Symbol	Description	Value	Units	Source
$T^{\mathrm{Gr}}_{\mathrm{surface}}$	Greenland regional surface temperature anomaly, averaged over the Berkeley Earth land cells in Greenland	—	°C	*Rohde and Hausfather* (2020)
ξ	Gain of the surface-to-ocean transfer function, the ratio of slow changes in subsurface ocean temperature to slow changes in Greenland surface temperature	fitted, 0.30 [0.16, 0.55]	—	this study
χ	Intercept of the surface-to-ocean transfer function	fitted, 0.05 [−0.08, 0.26]	°C	this study
τ_{go}	Response time of the subsurface ocean in the transfer function, weighted by likelihood over whole years from 1 to 20	fitted, 5 [3, 16]	yr	this study
$\varepsilon_{\mathrm{go}}$	Zero-mean residual of the transfer function	—	°C	—
AA	Arctic amplification factor, the ratio of Greenland to global warming, used to convert GMST to Greenland regional temperature for the discharge model.	2.5 ± 0.5	—	*Rantanen et al.* (2022)
Forecast construction				
$\dot{H}_{\mathrm{forecast}}$	Blended forecast rate of GMSL rise	—	$\mathrm{mm\,yr^{-1}}$	—
H_{forecast}	Forecast GMSL level	—	mm	—
$\dot{H}_{\mathrm{obs}}$	Rate from the satellite-era quadratic fit to altimetry	—	$\mathrm{mm\,yr^{-1}}$	*Hamlington et al.* (2024); *Nerem et al.* (2022)
$\dot{H}_{\mathrm{comp}}$	Rate of the summed calibrated components	—	$\mathrm{mm\,yr^{-1}}$	—
H_k	Level-space projection ensemble for component k	—	mm	—
w	Weight blending the observed and component rates	—	—	—
Θ	Standard logistic function	—	—	—
t_{r}	Midpoint year of the blend transition	2035 forecast, 2015 hindcast	yr	this study
τ_{r}	Width of the blend transition	5	yr	this study
Diagnostics				
$\dot{H}$	Rate of GMSL rise	—	$\mathrm{mm\,yr^{-1}}$	—
α	Transient sensitivity of the rate of GMSL rise to warming, $d\dot{H}/dT$	—	$\mathrm{mm\,yr^{-1}\ {}^{\circ}C^{-1}}$	—
R^2	Coefficient of determination	—	—	—

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Symbol	Description	Value	Units	Source
χ^2/dof	Reduced chi-square, the error-weighted sum of squared residuals per degree of freedom	—	—	—

1.3 Component model posteriors

This section collects the parameter posteriors of the calibrated rate–temperature and discharge models (Figs. S1–S5). Greenland discharge is fitted by weighted least squares to the annual increments of its cumulative record over a discrete set of delays, so its posterior is a mixture of normals weighted by the Bayesian Information Criterion and is shown as marginal densities. The remaining components are sampled by Markov chain Monte Carlo and are shown as corner plots, with marginal histograms on the diagonal and pairwise joint densities on the off-diagonal. Posterior medians and their 90% credible intervals are reported in the main text. The projections carry the full posterior draws.

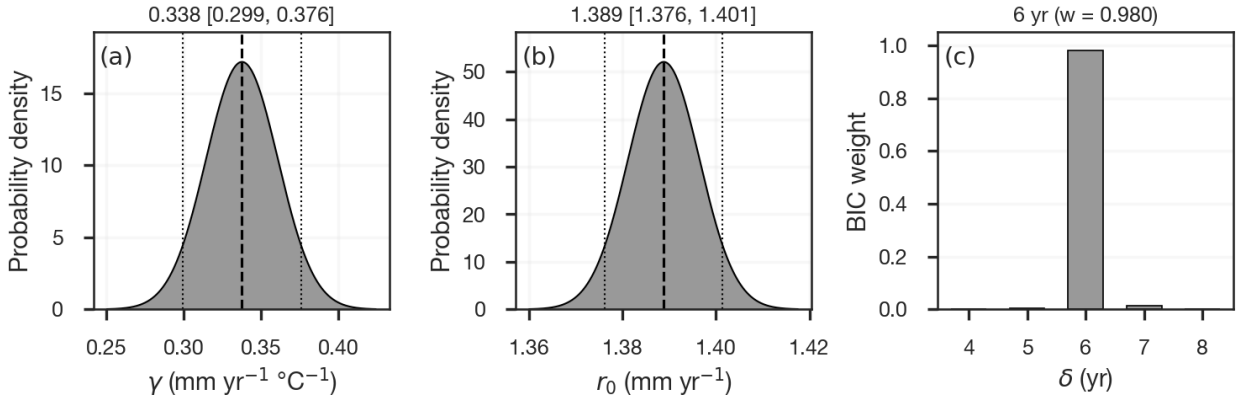


Figure S1: **Greenland discharge delay model parameter posteriors.** Marginal posteriors for the two fitted parameters and the selected delay of the discharge delay model calibrated against *Mouginot et al.* (2019). **(a)** The sensitivity of the discharge rate to the delayed ocean temperature anomaly, γ, and **(b)** the background discharge rate, r_0. **(c)** The delay δ is resolved only to integer years, so its posterior is the set of weights over the candidate delays, which place 98.0% of the weight on $\delta = 6\,\mathrm{yr}$. Panel titles give the median with the 90% credible interval.

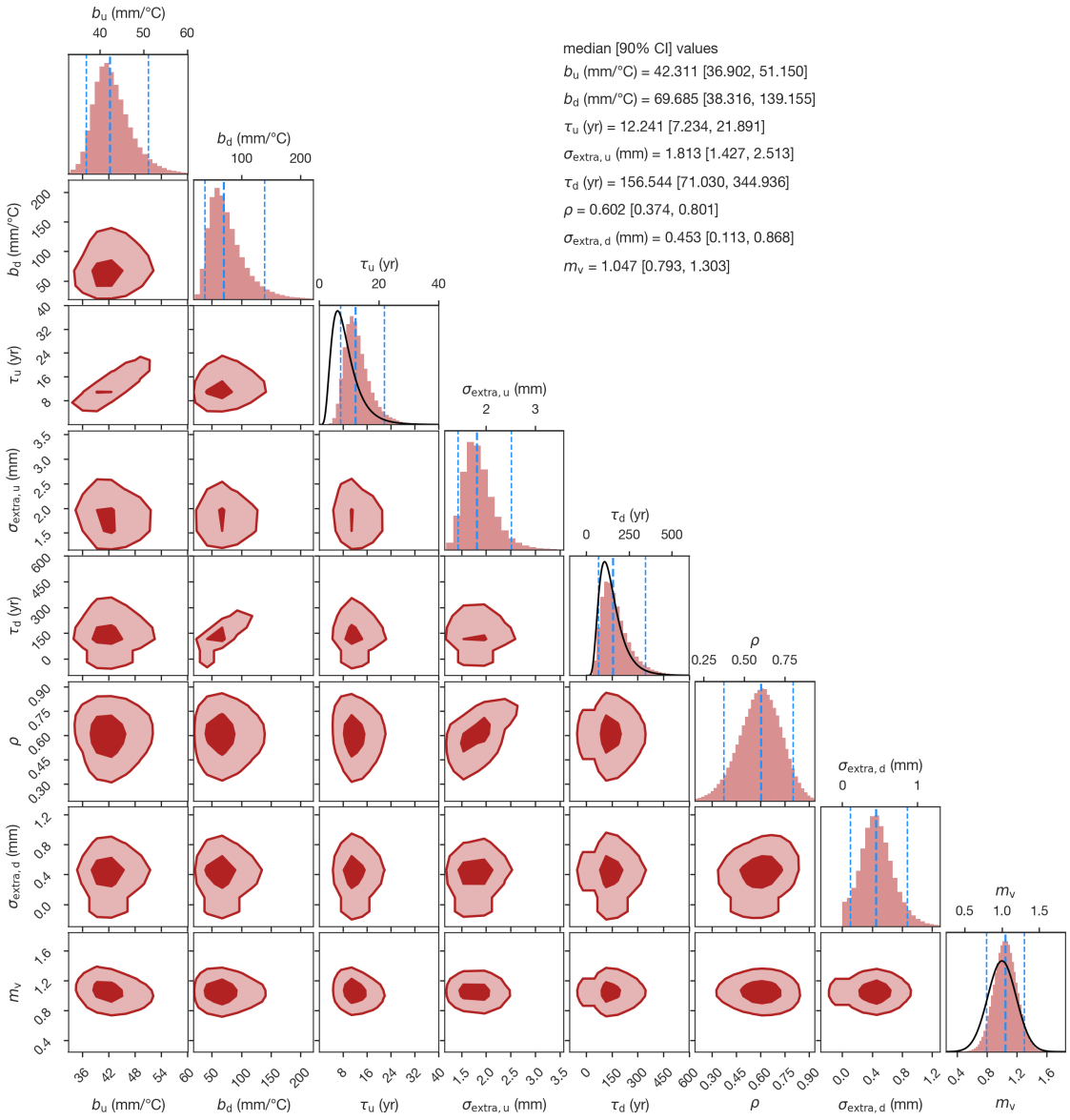


Figure S2: **Two-layer thermosteric model posterior correlation structure.** The joint posterior for the eight sampled parameters of the two-layer model fitted to NOAA 0–700 m and 700–2000 m thermosteric sea level data. Diagonal panels show marginal posterior histograms (red), with dashed vertical lines marking the median and 90% credible interval. Off-diagonal panels show the joint posterior density, with contours denoting 1 and 2 standard deviations. Black curves show the three informative priors, on τ_u, τ_d, and the volcanic multiplier m_v, while b_u, b_d, $\sigma_{extra,u}$, $\sigma_{extra,d}$, and ρ have flat priors that are not shown.

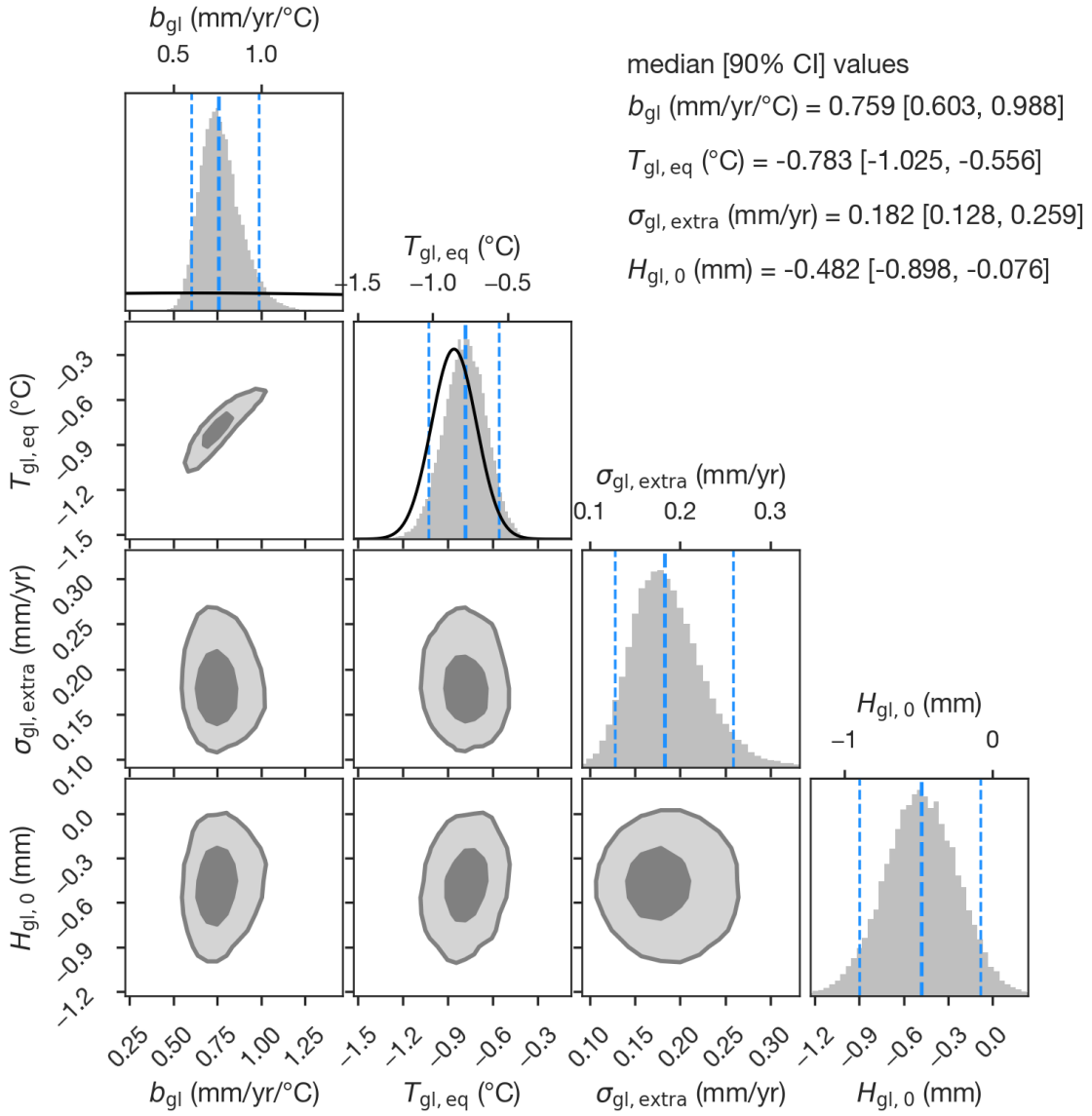


Figure S3: **Glacier model posterior correlation structure.** The joint posterior for the four free parameters of the glacier model (b_{gl}, $T_{gl,eq}$, $\sigma_{gl,extra}$, and $H_{gl,0}$) fitted to the GlaMBIE record (*The GlaMBIE Team et al.*, 2025), with the full covariance of the cumulative record in the likelihood. Diagonal panels show marginal posterior histograms (gray), with dashed vertical lines marking the median and 90% credible interval. Off-diagonal panels show the joint posterior density, with contours denoting 1 and 2 standard deviations. Black curves show the two priors: b_{gl}, which is centered on the sensitivity of *Rounce et al.* (2023) and weakly informative, and $T_{gl,eq}$, which is centered on the 1850–1879 mean GMST and fixes the rate at the 1995–2005 mean GMST, $c_{gl} = -b_{gl}T_{gl,eq}$, while $\sigma_{gl,extra}$ and $H_{gl,0}$ have wide priors that are not shown.

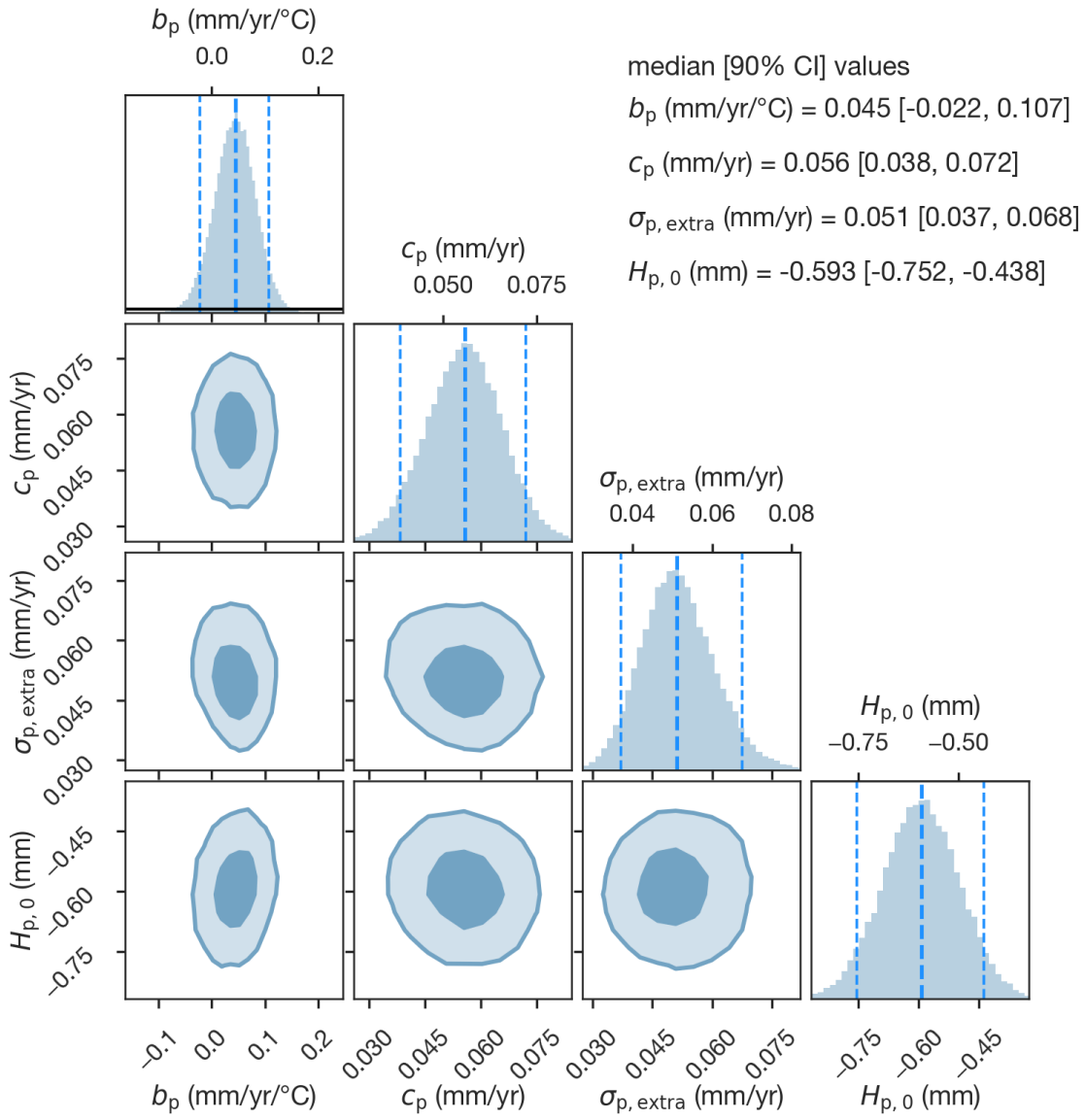


Figure S4: **Antarctic Peninsula model posterior correlation structure.** The joint posterior for the four parameters of the Peninsula model fitted to the IMBIE-3 Peninsula record (*Otosaka et al.*, 2026), with the full covariance of the cumulative record in the likelihood. The third parameter is the model-inadequacy term $\sigma_{p,extra}$, fitted here, for glaciers, and for WAIS, but not for EAIS. Diagonal panels show marginal posterior histograms (blue), with dashed vertical lines marking the median and 90% credible interval. Off-diagonal panels show the joint posterior density, with contours denoting 1 and 2 standard deviations. Only b_p has a prior shown (black curve), a symmetric normal that is flat over the range the data support, while the remaining three parameters have wide priors that are not shown.

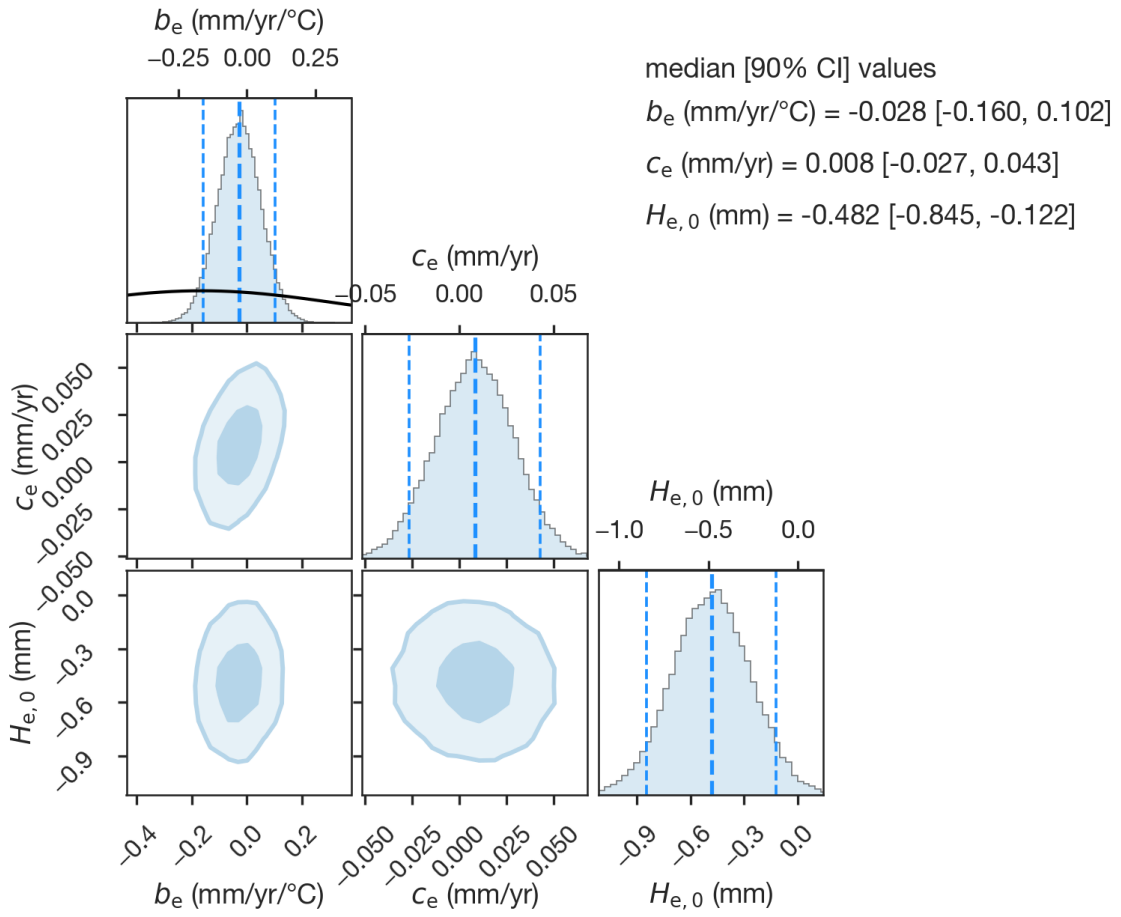


Figure S5: **EAIS model posterior correlation structure.** The joint posterior for the three parameters of the East Antarctic model fitted to the IMBIE-3 East Antarctic record (*Otosaka et al.*, 2026), with the full covariance of the cumulative record in the likelihood. This fit serves as a diagnostic, the projection instead adopting a literature surface-mass-balance sensitivity. Diagonal panels show marginal posterior histograms (pale blue), with dashed vertical lines marking the median and 90% credible interval. Off-diagonal panels show the joint posterior density, with contours denoting 1 and 2 standard deviations. Only b_e has a prior shown (black curve), a weakly informative normal prior centered on the literature sensitivity $C_T = 60$ Gt yr^{-1} °C^{-1} (*Frieler et al.*, 2015; *Ligtenberg et al.*, 2013) converted to sea level, with a standard deviation about nine times the literature uncertainty of ± 20 Gt yr^{-1} °C^{-1}, so the data set the posterior. The parameters c_e and $H_{e,0}$ have wide priors that are not shown.

1.4 Greenland surface mass balance emulator

We project Greenland SMB with a statistical emulator of the regional climate model MARv3.12 driven by seven CMIP6 global climate models (*Boberg et al.*, 2026): CESM2, MPI-ESM1-2-HR, NorESM2-MM, UKESM1-0-LL, CNRM-CM6-1, CNRM-ESM2-1, and IPSL-CM6A-LR. For each GCM, we fit the annual SMB contribution of the ice sheet as a quadratic in that GCM's own GMST anomaly T, relative to 1995–2005 and smoothed with an 11-year centered mean,

$$\dot{H}_{\mathrm{smb}} = a_{\mathrm{gris}}\, T^2 + b_{\mathrm{gris}}\, T + c_{\mathrm{gris}}, \tag{1}$$

with $\dot{H}_{\mathrm{smb}}$ in $\mathrm{mm\,SLE\,yr^{-1}}$ and positive for sea-level rise. Each fit pools the history run (1980–2014) with every SSP run available for that GCM (2015–2100). Each run is a separate segment with first-order autoregressive residuals. The posterior of a_{gris}, b_{gris}, and the intercept is analytic under a weak normal-inverse-gamma prior, and the autocorrelation is integrated over a grid weighted by marginal likelihood. We tested adding a lag between warming and the SMB response for CESM2 and MPI-ESM1-2-HR, the two GCMs with all three scenarios. It did not improve the fits, so the emulator has none.

In projections, we replace the intercept fitted to MAR with a level c_{gris} and a drift state $s(t)$, both fitted to the *Mouginot et al.* (2019) SMB. The emulator supplies the change in SMB with warming, and the observations supply its present level and its slow departures from the emulator. For each member, the residual of the observed SMB about that member's emulator response is modeled as $c_{\mathrm{gris}} + s(t) + \varepsilon(t)$, where s is a first-order autoregressive process with relaxation timescale τ_{s} and standard deviation σ_s, and ε combines year-to-year weather, with standard deviation σ_ε, and the reported observational error. The hyperparameters $(\tau_{\mathrm{s}}, \sigma_s, \sigma_\varepsilon)$ are estimated on a grid from the Kalman-filter likelihood of the residual about the median emulator, with a lognormal prior on τ_{s} whose 90% range is 5–25 yr and a weak half-normal prior on σ_s. Each member draws $(\tau_{\mathrm{s}}, \sigma_s, \sigma_\varepsilon)$ from this posterior and one joint path of c_{gris} and s through 1972–2018 by forward filtering and backward sampling. After 2018 the state follows its autoregressive process and relaxes toward zero, and before 1972 it is continued backward in the same way. The projections add white weather noise with the member's σ_ε and the SMB–elevation feedback of *Fettweis et al.* (2013). That feedback is an additional loss equal to a fraction $\epsilon_{\mathrm{fb}} \sim \mathcal{N}(0.08, 0.05^2)$, with negative draws set to zero, of the loss at their reference simulation, scaled by the member's cumulative SMB anomaly since 2000. Each member also draws one GCM, with equal weight, and one posterior draw of a_{gris} and b_{gris} for that GCM. Because MAR is the only regional climate model in the ensemble, the spread between regional climate models is not sampled. Under the same forcing, RACMO gives less SMB loss than MAR (*Glaude et al.*, 2024).

We test the emulator out of sample by leaving out each GCM in turn and predicting its runs from the other six (Fig. S7). The prediction is anchored to the withheld run's own 1995–2005 mean. Over 2015–2100, all 13 withheld runs fall within the 90% predictive range. The differences between MAR and the emulator range from −19.6 to +28.7 mm SLE, with a median absolute difference of 14.3 mm. The largest are for UKESM1-0-LL, at +17.4 mm (SSP2-4.5) and +28.7 mm (SSP5-8.5), where the other six GCMs underpredict its mass loss. Fig. S8 shows the same comparison year by year. The emulator is trained on MAR alone, so it does not sample the differences between regional climate models. Forced by the same CESM2 SSP5-8.5 run, RACMO gives an SMB contribution of 87 mm over 2015–2099 and

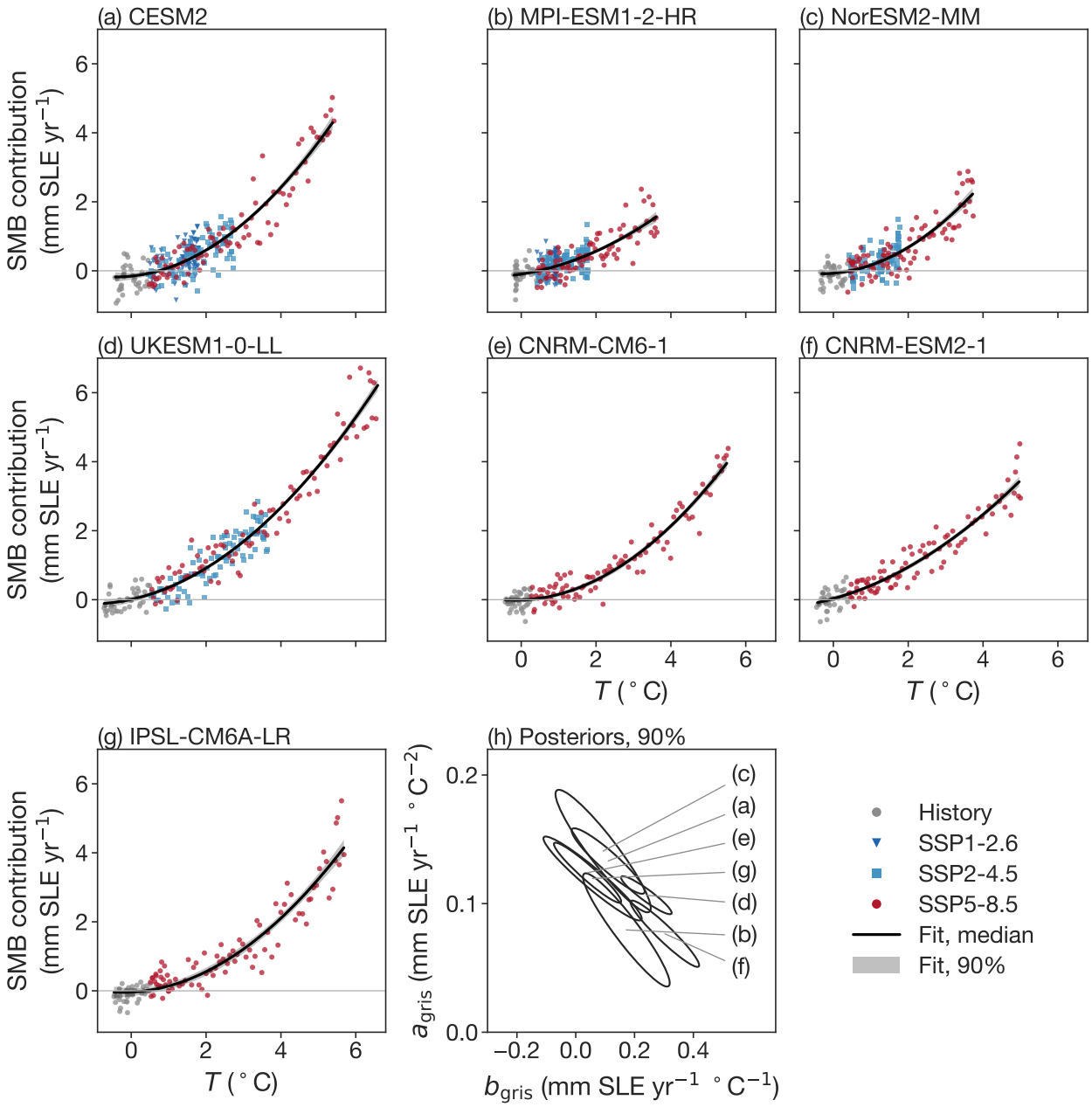


Figure S6: **Greenland SMB emulator fits.** **(a–g)** Annual SMB contribution of the contiguous Greenland ice sheet simulated by MARv3.12 driven by each of seven CMIP6 GCMs, relative to the 1995–2005 mean of that GCM's history run, against the GCM's own GMST anomaly T (11-year centered mean, relative to 1995–2005). Points are colored and shaped by experiment. Black curves show the posterior median of the fitted quadratic, including the fitted intercept, and gray bands its 90% credible interval. **(h)** 90% ellipse of the joint posterior of a_{gris} and b_{gris} for each GCM, labeled with the letter of its panel. All quantities are contributions to GMSL.

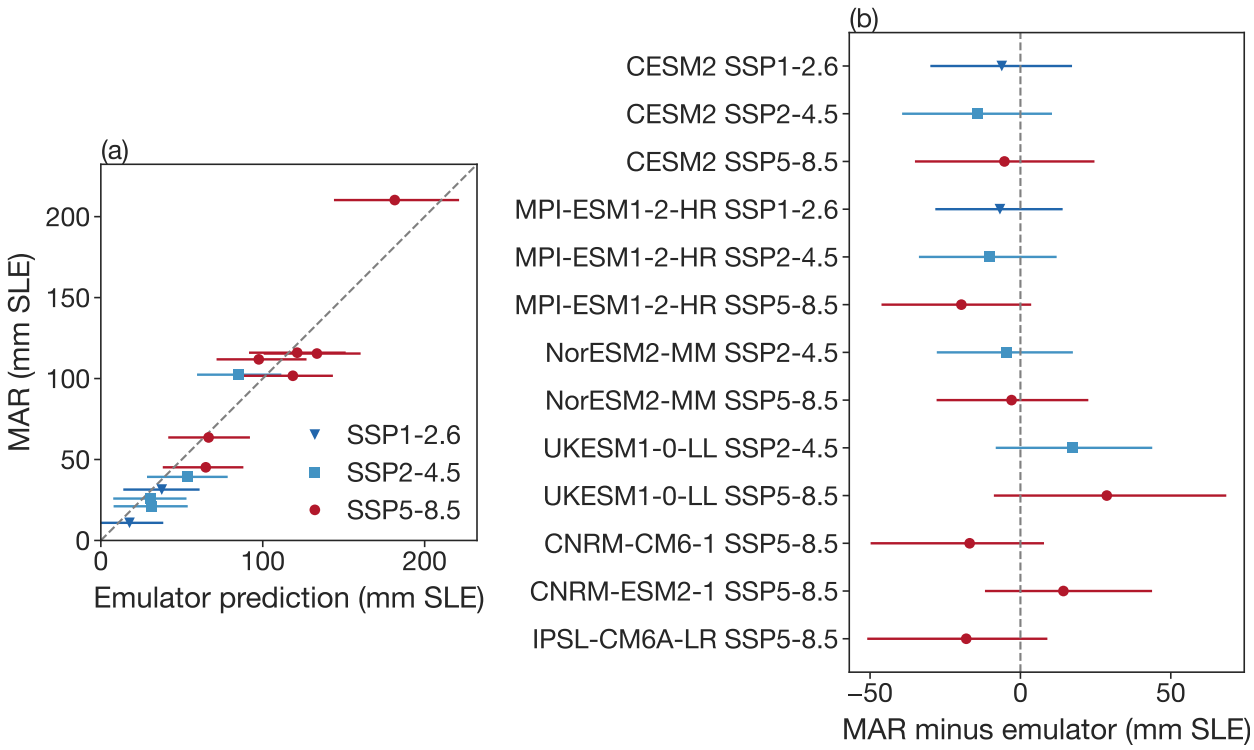


Figure S7: **Leave-one-GCM-out test of the Greenland SMB emulator.** Each GCM is withheld in turn, and its SSP runs are predicted by the emulator fitted to the other six GCMs, driven by the withheld GCM's GMST and anchored to its own 1995–2005 mean. **(a)** Cumulative SMB contribution over 2015–2100 relative to the 1995–2005 rate, MAR against the emulator. Horizontal bars show the 90% predictive range, which includes the spread of the six GCMs, the coefficient uncertainty, autoregressive weather noise, and the weather noise of the 1995–2005 anchor. The dashed line is the 1:1 line. **(b)** MAR minus the emulator median for each withheld run, with the same 90% range. Positive values mean that the emulator underpredicts the contribution to sea-level rise.

MAR gives 145 mm, against an emulator prediction of 117 [88, 146] mm. The two regional models differ by 58 mm, as much as the width of the emulator's 90% range (Fig. S8n).

Driven by observed GMST, the emulator's forced response declines more slowly than the observed SMB (Fig. S9). Its median SMB falls from 388 Gt yr^{-1} over 1972–1985 to 357 Gt yr^{-1} over 2010–2018, while the *Mouginot et al.* (2019) SMB falls from 466 to 252 Gt yr^{-1}. Every Mouginot year lies within the emulator's 90% range once weather noise is included, and 3 of the 38 years of the withheld *Mankoff et al.* (2021) SMB lie outside it (2010, 2012, and 2019). The emulator represents the response of SMB to global warming. It does not reproduce multi-year departures from that response, such as the low SMB of 2000–2018.

The drift state described above accounts for these departures. Fitted to the *Mouginot et al.* (2019) SMB over 1972–2018, it raises the maximum log likelihood from −293.4 for a constant level to −284.2. The likelihood is nearly flat in τ_s from about 10 to more than 100 yr, so τ_s follows its prior. Out of sample, the drift state helps little. Fitted to 1972–

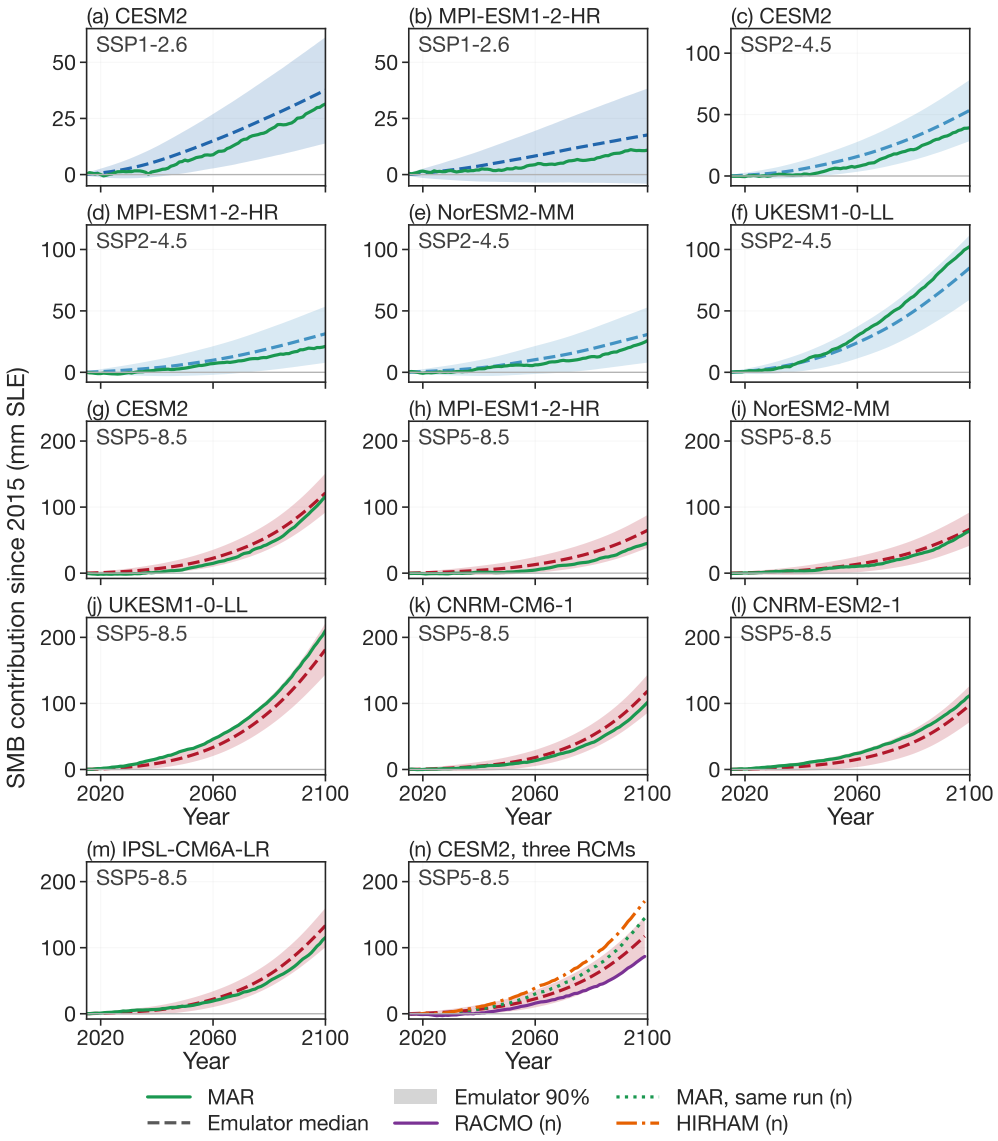


Figure S8: **Greenland SMB projections of regional climate models against the emulator, 2015–2100. (a–m)** Cumulative SMB contribution since 2015 of each MARv3.12 scenario run (*Boberg et al.*, 2026) (green), relative to the 1995–2005 mean of that GCM's history run, with the prediction of the emulator fitted to the other six GCMs and driven by the withheld GCM's GMST (dashed median and shaded 90% range, colored by scenario). The range includes the spread of the six GCMs, the coefficient uncertainty, autoregressive weather noise, and the weather noise of the 1995–2005 anchor, and the values at 2100 agree with those of Fig. S7 to within 1 mm, the Monte Carlo sampling noise. **(n)** RACMO2.3p2 (*Noël et al.*, 2021) (purple) and MAR (green dotted), both forced by a CESM2 SSP5-8.5 branch run, and HIRHAM (orange dash-dot), forced by CESM2 r1i1p1f1 under SSP5-8.5, all from *Glaude et al.* (2024), integrated over the contiguous ice sheet and taken relative to their own 1995–2005 means. RACMO ends in 2099. The emulator in (n) is fitted to the six GCMs other than CESM2 and driven by the CMIP6 CESM2 SSP5-8.5 GMST, because the GMST of the branch run is not archived. All quantities are contributions to GMSL.

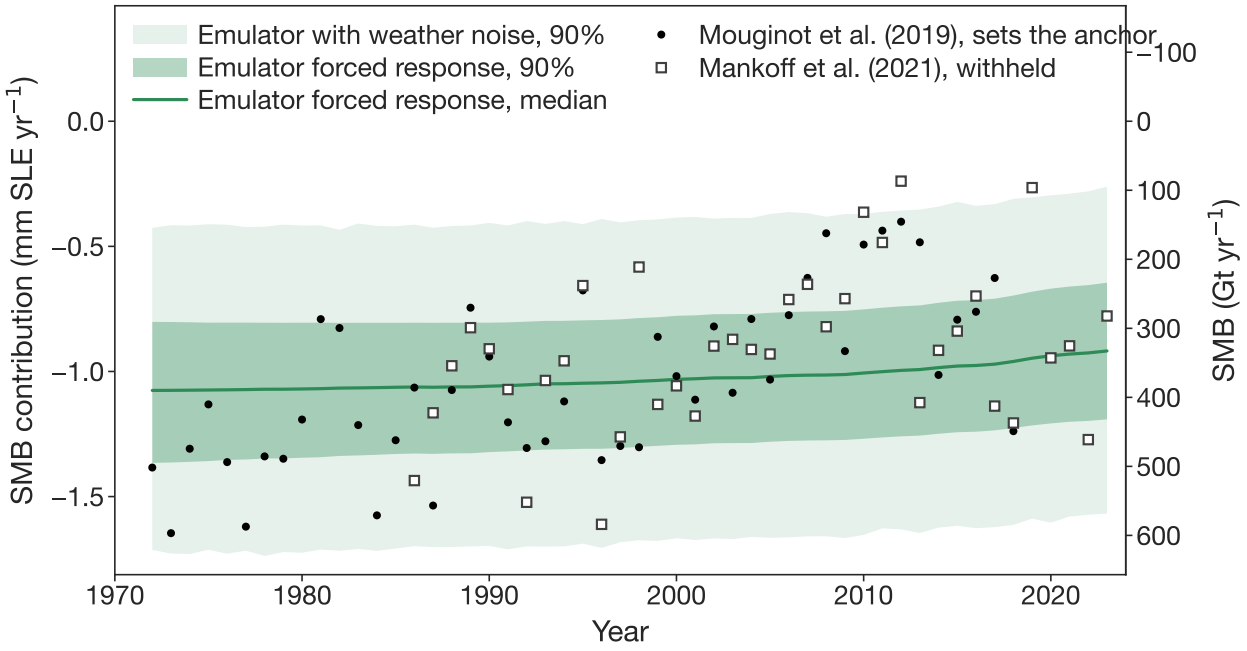


Figure S9: **Greenland SMB emulator driven by observed GMST.** The emulator's forced response (dark band, 90% range, and median), driven by the Berkeley Earth GMST anomaly (*Rohde and Hausfather*, 2020) with the 11-year centered mean used in training, is compared with the *Mouginot et al.* (2019) SMB, whose 1995–2005 mean anchors the emulator in this comparison, and the withheld *Mankoff et al.* (2021) SMB. The forced-response range includes the GCM, the coefficients, and the uncertainty of that anchor. The drift state is not included. The light band adds autoregressive weather noise with the parameters fitted to each GCM. The centered mean for 2020–2024 uses the IPCC AR6 2.7 °C GMST path (*Lee et al.*, 2021) for 2025–2029. The right axis gives the same SMB in Gt yr^{-1}.

2004, it forecasts the 2005–2018 SMB no better than a fixed 1995–2004 anchor, and both overpredict it. Fitted to 1972–2018, its forecast of the 2019–2023 mean SMB is slightly closer than the fixed anchor to the *Mankoff et al.* (2021) and IMBIE-3 (*Otosaka et al.*, 2026) records, and both again overpredict it. At 2100 the drift state changes the median SMB contribution by 2–3 mm and widens its 90% interval by 13–17 mm at 1.8–3.6 °C, mostly through the uncertainty of c_{gris}. A prior on τ_{s} that is uniform in $\log \tau_{\mathrm{s}}$ over 3–150 yr moves the 2100 medians by about 2 mm. Continued backward before 1972, the state adds 6 mm of SMB mass gain over 1900–1971 relative to a fixed anchor, because the record begins in a period of high SMB. This moves our Greenland hindcast further from the *Frederikse et al.* (2020) estimate, whose Greenland term before 1972 is the reconstruction of *Kjeldsen et al.* (2015).

1.5 Variance decomposition

We decompose the variance of the 2100 forecast separately for each warming trajectory. Pooling the trajectories would require assigning each one a probability. The total here is

the sum of the component models before blending with the observed trend, excluding terrestrial water storage. Under the fast-WAIS weighting, WAIS accounts for 99.2%, 98.8%, and 98.2% of the total variance at 1.8, 2.7, and 3.6 °C. The thermosteric, glacier, and Greenland contributions together account for 0.7%, 1.3%, and 2.1%, a share that includes the covariance among them from the warming path they share. EAIS and the Antarctic Peninsula account for the small remainder. WAIS does not depend on the warming trajectory or on the parameters of any other component, so these shares nearly sum to one. Under the slow-WAIS weighting, $p(\mathrm{S1}) = 0.90$, the WAIS variance is that of the two-branch mixture, $p\,\mathrm{var}(S_1) + (1-p)\,\mathrm{var}(S_2) + p(1-p)\,(\bar{S}_1 - \bar{S}_2)^2$, and WAIS accounts for 95.9%, 93.2%, and 89.4% of the total variance at 1.8, 2.7, and 3.6 °C.

1.6 Temperature scenarios

The component projections are driven by the IPCC AR6 projections of global mean surface temperature for each SSP (*Lee et al.*, 2021). As in the main text, we label each scenario by the AR6 central estimate of its warming over 2081–2100 relative to 1850–1900 (*Lee et al.*, 2021, Table 4.5): 1.4 °C (SSP1-1.9), 1.8 °C (SSP1-2.6), 2.7 °C (SSP2-4.5), 3.6 °C (SSP3-7.0), and 4.4 °C (SSP5-8.5). Each SSP has a median trajectory and a 5–95% range, which comes mostly from uncertainty in the response of the climate to a given emissions pathway. A world that warms faster than the median tends to stay warmer for the whole century. We therefore treat this range as a set of warming paths rather than as independent noise in each year.

Each ensemble member k is assigned one value z_k drawn from a standard normal distribution. Its temperature is the median trajectory plus an offset $\delta_k(t) = z_k\,[h(t) - h(t_0)]$ for $t > t_0$, and zero before t_0. When $z_k > 0$, $h(t)$ is the distance from the median to the 95th percentile divided by 1.645. When $z_k < 0$, it is the distance from the 5th percentile to the median divided by 1.645. This keeps the asymmetry of the AR6 range. We set $t_0 = 2024$ because warming up to the present is observed, so the offset adds spread only to future warming. As a result, the spread at 2100 is somewhat narrower than the full AR6 range. For the 2.7 °C scenario the upper half-width at 2099 is 0.64 °C rather than 0.82 °C.

Member k uses the same z_k in every component and every SSP. Each member therefore represents one world, in which the ocean, the glaciers, and the ice sheets all respond to the same warming. The z_k are drawn with their own random seed, so the parameter draws of each component do not change. Sampling the warming range widens the 2100 intervals of the temperature-driven components, most strongly for thermal expansion. At 2.7 °C its 90% interval is 70 mm wide, compared with 45 mm when the AR6 range instead enters as independent noise in each year.

1.7 Component-level skill metrics

Table S3 collects the goodness-of-fit statistics for every component fitted to data. We report reduced chi-square rather than a coefficient of determination. Every component is calibrated against a cumulative record whose variance is dominated by its trend, so R^2 in that space approaches unity for any model that reproduces the trend at all and says little about the

quality of the fit. A straight line in time, with no temperature dependence whatsoever, scores 0.982 against the 0–2000 m thermosteric record.

Reduced chi-square compares the residuals with the uncertainty the calibration data are reported to carry, so a value near unity indicates both that the model reproduces the observations to within their stated errors and that the error model used in the likelihood is well specified. The normalization is not identical in every case, and we note where it differs. Glaciers, the Antarctic Peninsula, EAIS, and WAIS are fitted to cumulative records whose errors are serially correlated, and their χ^2 is therefore evaluated against the full covariance of the cumulative record built from the reported annual uncertainties, not against a diagonal approximation, which would understate the residual scatter that the reported errors can absorb. Their degrees of freedom are the number of annual values minus all fitted parameters, including the initial level and, where fitted, the model-inadequacy term. All four sit between 0.90 and 1.09. Glaciers and the Peninsula reach those values only with fitted model-inadequacy terms, $\sigma_{\mathrm{gl,extra}}$ and $\sigma_{\mathrm{p,extra}}$, described in the main text. Without them, the reduced chi-square is 4.1 for glaciers, with the GlaMBIE 95% uncertainties converted to standard deviations, and 2.63 for the Peninsula. Both records therefore carry real annual scatter beyond their reported per-year uncertainties. EAIS does not need a model-inadequacy term. For the thermosteric model the quoted value uses the variance of each point, its observational error plus the fitted noise term $\sigma_{\mathrm{extra,u}}$ or $\sigma_{\mathrm{extra,d}}$, and ignores the autoregressive correlation of the noise terms. It is normalized by the number of data points rather than by the degrees of freedom, so it sits slightly below the corresponding χ^2/dof. The Greenland discharge value of 1.45 is evaluated on the annual increments of the cumulative record, which is equivalent to using its full covariance, before the error scale is applied. It is the factor by which the parameter covariance is inflated, so it is a statement about the reported errors being too small rather than about the model missing the data. It has 38 degrees of freedom, the 40 annual increments between the 41 fitted points minus two for the fitted parameters; the error scale is not counted, because the value reported is the error scale itself. The *Mouginot et al.* (2019) discharge record runs from 1972, but we use EN4 from 1970 onward, so the delayed ocean forcing is undefined in the first years of the record. Every candidate lag is therefore fit to the same 41 years, 1978 to 2018, those covered by the longest candidate lag of 8 yr, and the first six years of the discharge record do not enter the fit.

Two components need further comment. WAIS is fitted in time rather than in temperature, but with the same annual-increment likelihood as the Peninsula, including a fitted model-inadequacy term, so its reduced chi-square of 1.01 is comparable with the other rows. A cubic in time is not supported by the record, with $\Delta\mathrm{BIC} = +3.6$ relative to the quadratic. Snowfall anomalies also modulate the record. In the Amundsen Sea Embayment, persistently low snowfall added 51 ± 4 Gt yr^{-1} to the mass loss during 2009–2013, and extreme snowfall reduced it by 60 ± 16 Gt yr^{-1} in 2019 and 2020 (*Davison et al.*, 2023). To test their effect on the fit, we removed both anomalies from the IMBIE-3 record, converting at 362.5 Gt per mm SLE, and refitted the S1 model with the same likelihood. With the anomalies removed, the contribution at 2100 relative to 2000 falls from 86 to 79 mm. This test is approximate, because *Davison et al.* (2023) report the anomalies for the Amundsen Sea Embayment alone and as multi-year means. The 7 mm change is small relative to the S1 90% interval of [−10, 182] mm, so we fit the record as reported.

We also tested curvature in the Peninsula model by fitting a quadratic rate model,

$\dot{H}_\mathrm{p} = a_\mathrm{p}T^2 + b_\mathrm{p}T + c_\mathrm{p}$, with a_p unrestricted in sign. With the BIC computed from the maximized likelihood, including the fitted $\sigma_\mathrm{p,extra}$, the quadratic is preferred ($\Delta\mathrm{BIC} = 4.6$), with negative curvature, $a_\mathrm{p} = -0.35$ [-0.54, -0.16] mm yr^{-1} °C^{-2}. This preference comes from the warmest years of the record. The Peninsula loss rate averaged 0.12 mm yr^{-1} over 2005–2014 and 0.01 mm yr^{-1} over 2015–2023, and in the IMBIE-3 partition about half of this decline comes from increased surface mass balance and half from slower ice discharge (*Otosaka et al.*, 2026). Fitted to 1979–2019 alone, the quadratic is no longer clearly preferred ($\Delta\mathrm{BIC} = 0.3$). Extrapolated, the quadratic would make the Peninsula gain mass whenever GMST is more than 0.6 °C above its 1995–2005 mean, a level reached in 2023, and at 3 °C above that mean the gain would be about 1000 Gt yr^{-1}. A recent surface-mass-balance anomaly and a slowdown in discharge do not support a response that grows with the square of warming, so we keep the linear model and set $a_\mathrm{p} = 0$.

EAIS is listed as a diagnostic because the East Antarctic Ice Sheet is close to balance over the IMBIE-3 record, leaving little signal for a temperature-dependent model to recover. The fit is consistent with the reported errors, but the projection adopts a literature surface-mass-balance sensitivity, $C_T = 60 \pm 20$ Gt yr^{-1} °C^{-1} (*Frieler et al.*, 2015; *Ligtenberg et al.*, 2013), and uses the regression only as a check on that value.

One further fit enters the framework without being a component model in its own right. The surface-to-ocean transfer function given in the main text carries SSP temperature trajectories into the Greenland discharge model. The quantity it must supply is the multi-decadal trend in subsurface temperature, which the ocean follows, rather than the year-to-year variability of surface temperature, which it largely does not, so it relates ocean temperature to a running average of surface temperature. At its most likely response time it explains $R^2 = 0.65$ of the variance of the 51 annual values, with a residual standard deviation of 0.15 °C that is propagated into the projections. Its gain is the least certain link in the Greenland projection chain, because the record constrains long response times only weakly. Greenland SMB is passed through as the RACMO-derived *Mouginot et al.* (2019) SMB over the historical period and projected with the emulator of MARv3.12 described in Section 1.4, whose response to warming is not fitted to observations. Its level and its slow departures from the emulator are fitted to the *Mouginot et al.* (2019) SMB through the drift state. The emulator's out-of-sample test is the leave-one-GCM-out comparison in Fig. S7, and the drift state's are the forecast tests in Section 1.4.

These are all within-sample statistics, and on their own they establish only that each model is consistent with the record it was fitted to. The tests that bear on forecasting skill are out of sample. The thermosteric model is compared against the *Frederikse et al.* (2020) steric reconstruction, withheld from calibration, with an RMS difference of 2.97 mm over 1957–2018 and 5.00 mm before 1957, where that reconstruction is itself constrained by models rather than by subsurface observations. When the thermosteric model is refitted with the 0–700 m record before 1990 withheld, retaining the full 700–2000 m record, which begins in 2005, it gives $b_\mathrm{u} = 43.0$ [36.7, 52.0] mm °C^{-1}, compared with 42.3 [36.9, 51.2] mm °C^{-1} from the full record. It predicts a rate of 0.22 [0.17, 0.29] mm yr^{-1} over the withheld years 1955–1989, compared with 0.20 [0.16, 0.24] mm yr^{-1} observed. In the prediction, the withheld years lie 12.6 [9.3, 16.4] mm below the 1995–2005 mean, compared with 12.0 [11.4, 12.6] mm observed, and 94% of the withheld annual values fall within the 90% predictive interval (Fig. S11). GMST rose by only 0.3 to 0.4 °C over the withheld years, so this test constrains

the level of the fit more than its sensitivity to warming. The glacier model is compared against the global glacier mass changes of *Zemp et al.* (2019) for 1962–1999, withheld from calibration, and gives a mean rate of 0.30 [0.17, 0.40] mm yr^{-1}, compared with an observed 0.32 [−0.02, 0.66] mm yr^{-1}. For this comparison the posterior rate $b_{\mathrm{gl}}T + c_{\mathrm{gl}}$ is evaluated for each hydrological year (October to September) with T weighted 0.25 from the preceding calendar year and 0.75 from the current one. The uncertainties of *Zemp et al.* (2019), reported as 95% intervals, are converted to standard deviations and combined following their prescription, with the glaciological, interpolation, and area terms added in quadrature across years and the geodetic term added linearly. Like the GlaMBIE global total, their record includes the Greenland and Antarctic peripheral glaciers. Fitted to 1962–1989 alone, the *Zemp et al.* (2019) record shows no detectable sensitivity to GMST, unlike 1990–2016. A delayed glacier response does not account for the difference: for response times of 10 to 100 years the two sensitivities remain 2.5 to 3.9 standard deviations apart. A quadratic dependence on GMST fitted to 1962–2016 reduces the separation from 3.5 to 1.5 standard deviations but fits the record less well than separate linear fits to the two periods. Greenland discharge is compared against the *Mankoff et al.* (2021) record (1986–2023) and the IMBIE-3 time series (*Otosaka et al.*, 2026), neither of which enters the fit. The model falls within error of both for discharge and for total mass balance. As noted in the main text, the *Mankoff et al.* (2021) comparison is the weaker of the two, because that record shares its underlying raw data with the *Mouginot et al.* (2019) calibration and differs from it in processing rather than in provenance. IMBIE-3 is the more informative test. It combines mass-balance estimates from satellite observations of ice flow, surface elevation, and gravimetry. Its input-output estimates, which are built from ice flow, include *Mouginot et al.* (2019) and *Mankoff et al.* (2021). Its altimetry estimates (from 1992) and gravimetry estimates (from 2002) are independent of the calibration, so the independent part of the test is the total mass balance after 1992. When the discharge model is refitted to the 1990–2018 part of the *Mouginot et al.* (2019) record alone, it gives $\gamma = 0.345$ [0.296, 0.393] mm SLE yr^{-1} °C^{-1} and predicts 15.27 [15.06, 15.53] mm SLE of cumulative discharge over the withheld years 1978–1990, compared with 15.23 [14.96, 15.49] mm SLE observed. The surface-to-ocean transfer function is tested in the same way on the EN4 years before 1970. Over 1958–1969 the EN4 analysis in the Greenland box draws on at least 1,337 quality-controlled profiles a year, more than the 982 of the sparsest year of the fit. Fitted to 1970–2020 and driven by the observed Greenland regional surface temperature alone, the transfer function predicts a 1958–1969 mean of −0.12 [−0.32, 0.08] °C relative to 1995–2005, compared with −0.11 °C in EN4, and all 12 withheld annual values fall within the 90% predictive interval. The predicted mean changes by less than 0.03 °C across response times of 1 to 20 yr, so, like the thermosteric holdout, this test constrains the level of the fit rather than its response time. The tests for the framework as a whole rather than one component at a time are budget closure, in which the summed components are compared with satellite altimetry measurements that no component calibration saw, and the hindcast of the level, rate, and transient sensitivity of GMSL since 1950. Both are presented in the Results of the main text.

The hindcast depends on the TWS term before 2003, when the *Frederikse et al.* (2020) TWS is not constrained by gravimetry. Over 1950–2000, the reconstructed TWS lowers sea level, mainly through water impoundment in artificial reservoirs (*Frederikse et al.*, 2020). With it, the component sum rises at 0.75 mm yr^{-1} over 1950–2000. With TWS set to

zero before 2003 and the *Frederikse et al.* (2020) change since 2003 retained, it rises at 1.18 mm yr^{-1}, compared with 1.01 to 1.28 mm yr^{-1} in the GMSL records (Fig. S10). The GMSL reconstructions contain the same change in TWS, so the hindcast compared with them in the main text uses the full TWS record.

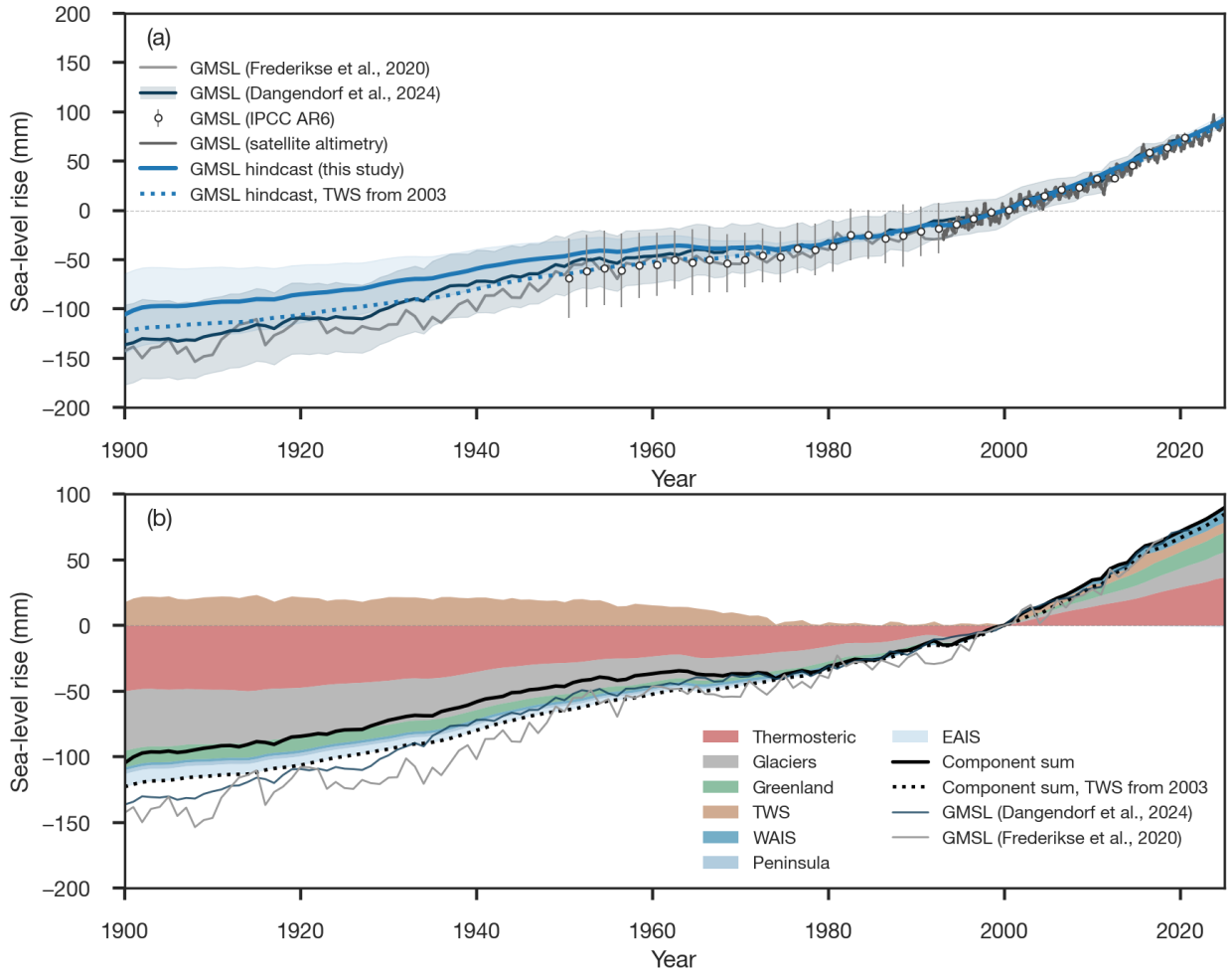


Figure S10: **Hindcast of GMSL and its dependence on terrestrial water storage before 2003. (a)** GMSL from *Frederikse et al.* (2020), *Dangendorf et al.* (2024) with its 90% interval, the IPCC AR6 compilation (*Fox-Kemper et al.*, 2021) with 90% intervals (every other year), and satellite altimetry, with the hindcast of Fig. 1a of the main text (median and 90% credible interval) and the median hindcast with TWS set to zero before 2003 (dotted). **(b)** Median contribution of each component to the component sum, stacked separately above and below zero, with the median component sum using the full TWS record (solid) and with TWS set to zero before 2003 (dotted), and the *Frederikse et al.* (2020) and *Dangendorf et al.* (2024) reconstructions. All series are relative to 2000.

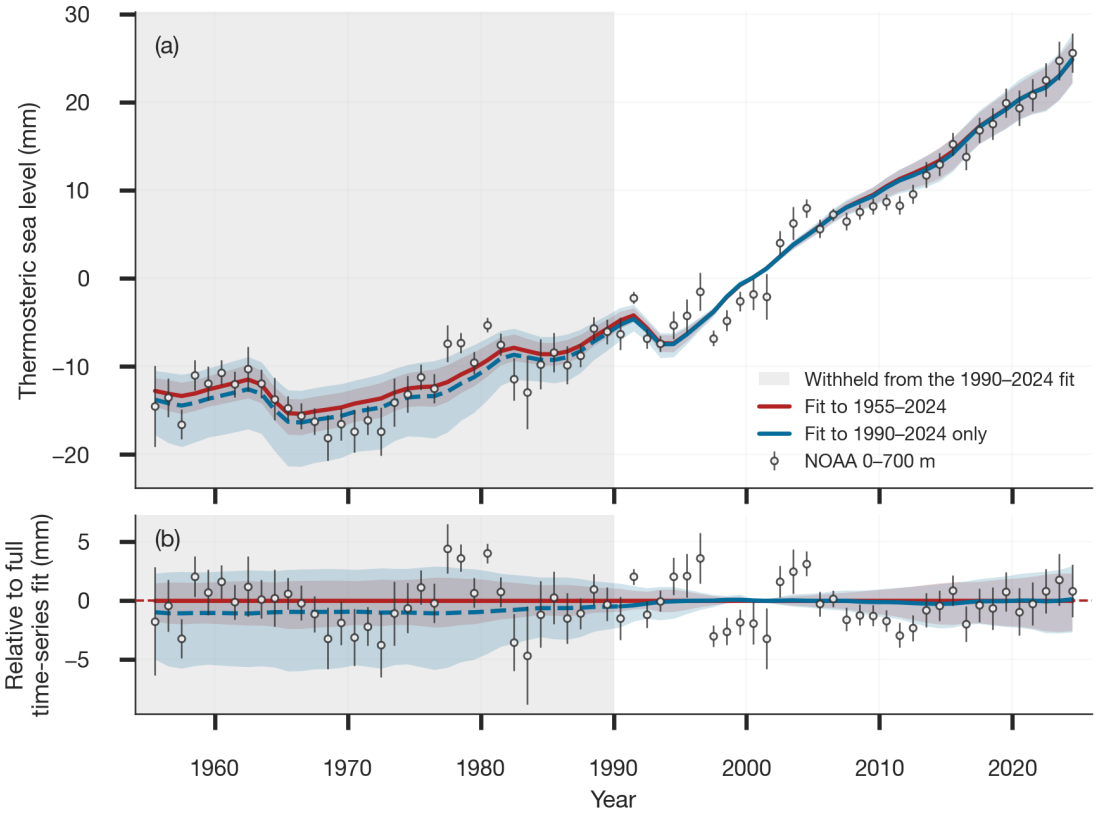


Figure S11: **Early-record holdout of the two-layer thermosteric model. (a)** NOAA 0–700 m thermosteric sea level (circles, with 90% intervals of the formal error and the 5% structural error; the fitted $\sigma_{\mathrm{extra,u}}$ is not included) with the model fitted to the full 1955–2024 record (red) and to 1990–2024 alone (blue), each shown as the posterior median and 90% credible interval. The model and the data are referenced to their 1995–2005 means, and the years withheld from the 1990–2024 fit are shaded. The 1990–2024 fit is dashed over the withheld years. **(b)** The same quantities relative to the median of the full time-series fit. The bands show parameter uncertainty only and do not include the fitted observation noise $\sigma_{\mathrm{extra,u}}$.

Table S3: Goodness-of-fit for the component models fitted to data. For glaciers, the Peninsula, EAIS, and WAIS, χ^2/dof is evaluated against the full covariance of the cumulative record, including the fitted model-inadequacy term where there is one, and the degrees of freedom count every fitted parameter. For the thermosteric model it is normalized by the number of data points rather than by the degrees of freedom and includes the fitted $\sigma_{\mathrm{extra,u}}$ or $\sigma_{\mathrm{extra,d}}$ in the point variance, without their autoregressive correlation. For Greenland discharge it is evaluated on the annual increments of the cumulative record, which is equivalent to using its full covariance, before the error scale is applied. Dataset references are given in Table S1. Greenland SMB, whose emulator is fitted to regional climate model output rather than to observations (Section 1.4), and terrestrial water storage are omitted.

Component	Calibration data	Reduced χ^2	Model
Thermosteric (0–700 m)	NOAA, 1955–2024 (70)	0.86	Two-layer, 8 sampled
Thermosteric (700–2000 m)	NOAA, 2005–2024 (20)	0.58	Joint with 0–700 m
Glaciers	GlaMBIE, 2000–2023 (24)	1.09	Linear rate–T, 1850 balance, σ_{extra}
Greenland discharge	Mouginot, 1978–2018 (41)	1.45	Delay, 2 fitted
Antarctic Peninsula	IMBIE-3, 1979–2023 (45)	1.00	Linear rate–T
EAIS (diagnostic)	IMBIE-3, 1979–2023 (45)	0.90	Linear rate–T
WAIS	IMBIE-3, 1979–2023 (45)	1.01	Linear rate in time, σ_{extra}

1.8 Energetic consistency of the thermosteric projections

The thermosteric model is driven by GMST and contains no explicit energy budget. We therefore check that its projections imply ocean heat uptake consistent with that of the scenarios that produce the warming. We convert the projected thermosteric rise of each ensemble member to cumulative ocean heat uptake by dividing it by the expansion efficiency of heat, the thermosteric rise per unit of heat added to the ocean (*Kuhlbrodt and Gregory*, 2012). We use the AR6 value of $0.113 \pm 0.013\,\mathrm{m\,YJ^{-1}}$ (mean and standard deviation; $1\,\mathrm{YJ} = 10^{24}\,\mathrm{J}$), with which AR6 converts the ocean heat uptake of its two-layer emulator to thermosteric sea level (*Fox-Kemper et al.*, 2021). We compare the change between the 1995–2014 and 2081–2100 means, the interval over which AR6 reports ocean heat content projections.

At the AR6 expansion efficiency, the implied heat uptake is 0.79 to 0.82 times the median of the AR6 two-layer emulator in the four scenarios (Table S4), consistent with our thermosteric projections lying below the AR6 medians (main text). Matching the AR6 medians would require an expansion efficiency of 0.090 to $0.093\,\mathrm{m\,YJ^{-1}}$, which is 1.5 to 1.8 standard deviations below the AR6 value. The implied heat uptake therefore falls short of the AR6 emulator by more than the uncertainty of the expansion efficiency. The heat uptake at 4.4 °C is 1.96 times that at 1.8 °C, compared with 1.89 for the AR6 emulator, so the gap narrows slightly with warming.

The upper 2000 m can be compared with CMIP6 directly. In the NOAA 0–2000 m record over 1955–2010, thermosteric sea level rose by 0.126 m per YJ of added heat (*Levitus et al.*, 2012). With this value, the implied heat uptake in the upper 2000 m is 0.72 to 0.83 times the CMIP6 median in the four scenarios (Table S4). This comparison is less direct than the full-depth one. Seawater expands less per unit of heat where it is cold, and the model places a larger share of the 0–2000 m rise below 700 m than the NOAA record showed over

1955–2010, so the appropriate expansion efficiency for the projections may be lower than the observed value.

Table S4: Ocean heat uptake implied by the thermosteric projections, as the change between the 1995–2014 and 2081–2100 means. Our values are medians with 90% credible intervals in brackets. Implied heat uptake is the thermosteric rise divided by the expansion efficiency of heat, either fixed at the AR6 value of 0.113 m YJ^{-1} or drawn for each ensemble member from a normal distribution with mean 0.113 and standard deviation 0.013 m YJ^{-1}. For 0–2000 m the expansion efficiency is 0.126 m YJ^{-1}, the ratio of thermosteric rise to heat content change in the NOAA 0–2000 m record over 1955–2010 (*Levitus et al.*, 2012). AR6 values are medians with 17–83% ranges in parentheses, for the two-layer emulator (full depth) and for CMIP6 models (0–2000 m) (*Fox-Kemper et al.*, 2021, Section 9.2.2.1 and Table 9.1).

	Full depth				0–2000 m	
Scenario	Thermosteric (mm)	Heat, fixed (YJ)	Heat, drawn (YJ)	AR6 two-layer (YJ)	Heat (YJ)	CMIP6 (YJ)
1.8 °C	106 [84, 132]	0.94 [0.75, 1.17]	0.94 [0.71, 1.27]	1.18 (0.99–1.42)	0.76 [0.60, 0.97]	1.06 (0.80–1.31)
2.7 °C	142 [116, 175]	1.26 [1.03, 1.55]	1.26 [0.97, 1.69]	1.56 (1.33–1.86)	1.05 [0.85, 1.31]	1.35 (1.08–1.67)
3.6 °C	176 [144, 214]	1.56 [1.28, 1.90]	1.56 [1.20, 2.08]	1.90 (1.63–2.21)	1.32 [1.07, 1.62]	1.62 (1.37–1.91)
4.4 °C	208 [165, 259]	1.84 [1.46, 2.29]	1.85 [1.39, 2.50]	2.23 (1.92–2.64)	1.57 [1.24, 1.98]	1.89 (1.60–2.29)

1.9 Sensitivity to the thermosteric calibration record

We test how much the thermosteric projections depend on the choice of calibration record by refitting the model with two other records in place of the NOAA records (Fig. S12). The first is the EN4.2.2 objective analysis of ocean temperature and salinity (*Good et al.*, 2013). From it we compute thermosteric sea level with the TEOS-10 equation of state (*IOC, SCOR and IAPSO*, 2010), holding salinity at its 1971–2000 mean. The second is the IAP v4.2 ocean heat content (*Cheng et al.*, 2017), which fills unsampled regions using covariances from CMIP5 historical simulations. We convert its heat content to thermosteric sea level with the ratio of the two in EN4 over 2005–2024, which is 0.128 mm per ZJ for 0–700 m and 0.114 mm per ZJ for 0–2000 m ($1\,\mathrm{ZJ} = 10^{21}\,\mathrm{J}$). This assumes that the IAP warming has the same spatial pattern as the EN4 warming. Neither product reports a standard error for its global mean, so we use the NOAA standard error of the same year. Each record is fitted over the same years as the NOAA records, both unscaled and with the factor of 1.14.

The three records differ most in the early decades and in the last decade (Fig. S12a,b). We compare their trends through the trend of the difference between two records, which removes the variability they share, and give 90% intervals that account for autocorrelated residuals. Over 1955–2012, the 0–700 m trend of IAP exceeds that of NOAA by 0.04 [0.00, 0.08] mm yr^{-1}, and the trend of EN4 falls below it by 0.11 [0.01, 0.22] mm yr^{-1}. EN4 is flatter than the other two records before about 1975, when sparse observations leave its analysis close to the climatology toward which it relaxes (*Good et al.*, 2013). Over 2013–2024, EN4 and IAP both rise faster than NOAA. Their trends exceed that of NOAA by 0.20 [0.05, 0.35] and 0.18 [0.02, 0.33] mm yr^{-1} over 0–700 m and by 0.29 [0.13, 0.44] and 0.17 [0.09, 0.26] mm yr^{-1} over 0–2000 m.

At equal scaling, calibrating with EN4 or IAP instead of NOAA raises the 2100 medians by 4 to 13 mm at 1.8, 2.7, and 3.6 °C (Fig. S12e). The factor of 1.14 raises them by 14 to 27 mm. Every median lies within the 90% credible interval of our projections. With the factor applied, the model fits the IAP record with a smaller noise term than the NOAA record, $\sigma_{\mathrm{extra,u}} = 1.46$ mm compared with 1.81 mm, and fits the EN4 record with a larger one, 3.21 mm, whose residuals are also more strongly autocorrelated ($\rho = 0.85$ compared with 0.60).

We also compare the records and our model with global-mean thermosteric sea level simulated by CMIP6 models (Fig. S12c,d). We use the full-depth thermosteric sea level of 30 models from the CMIP6 archive, 25 of which continue under SSP2-4.5 to 2024. One of these, GISS-E2-1-G, steps at the junction of its historical and SSP2-4.5 runs, so we use it only to 2014. From each model we remove the linear drift of its preindustrial control run over the years parallel to 1850–2024. The control run of GISS-E2-1-G-CC ends before these years do, so we fit its drift over the part that it covers. For UKESM1-1-LL, whose branch point lies outside the archived control run, we fit the drift over the whole control run. Over 2005–2024, the median full-depth trend across models is 1.50 mm yr^{-1}, with 5th and 95th percentiles of 0.99 and 2.00 mm yr^{-1}. This compares with 1.44 [1.28, 1.58] mm yr^{-1} for our model and 1.39 to 1.54 mm yr^{-1} for the unscaled 0–2000 m records plus the rate below 2000 m. Over 1955–2012, the median model trend of 0.38 mm yr^{-1} is lower than our full-depth trend of 0.72 [0.64, 0.80] mm yr^{-1} and close to our 0–700 m trend of 0.42 [0.38, 0.46] mm yr^{-1}. The difference therefore lies below 700 m, where the records constrain our model only from 2005 onward.

We also compare the full-depth rise per degree of warming, with each model scaled by its own GMST change. Between 1955–1974 and 2005–2024, our model gives 50 [45, 55] mm °C^{-1}, compared with a median of 33 mm °C^{-1} across models (5th and 95th percentiles 21 and 55 mm °C^{-1}). Between 1995–2014 and 2081–2100 under SSP2-4.5, our model gives 76 [65, 90] mm °C^{-1}, with each ensemble member scaled by its own warming path, close to the median of 78 mm °C^{-1} across models (69 and 112 mm °C^{-1}). The AR6 projections give 93 mm °C^{-1} as the ratio of their median thermal expansion to their median warming. They come from a two-layer emulator constrained by the assessed climate sensitivity, not directly from CMIP6 (*Fox-Kemper et al.*, 2021). Our historical rise per degree is thus higher than in most models, and our rise per degree grows less over the century than in the models or in AR6. In our model this growth is set by the deep layer, which the records constrain weakly.

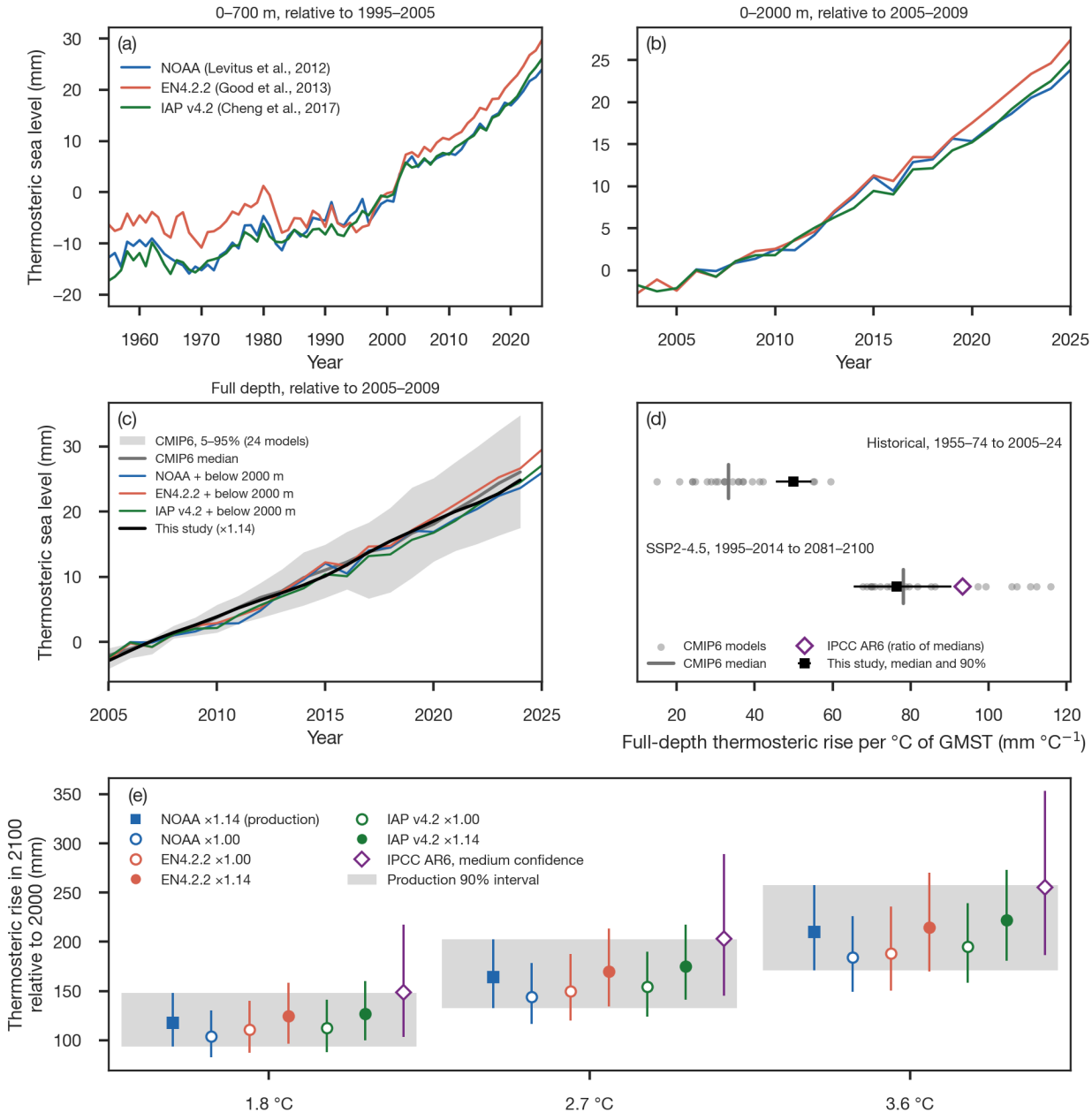


Figure S12: **Sensitivity of the thermosteric projections to the calibration record.** **(a)** Thermosteric sea level over 0–700 m from NOAA (*Levitus et al.*, 2012), EN4.2.2 (*Good et al.*, 2013), and IAP v4.2 (*Cheng et al.*, 2017), each relative to its 1995–2005 mean and without the factor of 1.14. **(b)** The same over 0–2000 m, relative to the 2005–2009 mean. IAP heat content is converted to thermosteric sea level with the ratio of the two in EN4 over 2005–2024. **(c)** Full-depth thermosteric sea level relative to the 2005–2009 mean. The gray band and line are the 5th–95th percentile range and median of 24 CMIP6 models (historical and SSP2-4.5, drift removed). Colored lines are the unscaled 0–2000 m records plus the rate below 2000 m (*Desbruyères et al.*, 2016), and the black line is the median of our model. **(d)** Full-depth thermosteric rise per °C of GMST between 1955–1974 and 2005–2024 (upper) and between 1995–2014 and 2081–2100 under SSP2-4.5 (lower). Gray dots are CMIP6 models, each scaled by its own warming, and vertical bars their median. Squares are the medians and 90% credible intervals of our model, scaled by Berkeley Earth GMST (*Rohde and Hausfather*, 2020) (upper) and by the warming path of each ensemble member (lower). The diamond is the ratio of the median AR6 thermal expansion to the median AR6 warming (*Fox-Kemper et al.*, 2021). Only the horizontal position carries a value. **(e)**

2 Exceedance Probability by Threshold and Scenario

Table S5 reports the probability of exceeding selected global mean sea-level thresholds (relative to pre-industrial) by 2100 under three warming trajectories, for the slow-WAIS case, $p(\mathrm{S1}) = 0.90$, and the fast-WAIS case, $p(\mathrm{S1}) = 0.10$. Together these columns isolate the contribution of WAIS tail risk: the difference between them is the effect of shifting the scenario weight from very unlikely to very likely continued mass loss along the observed trajectory. The slow-WAIS column retains 10% weight on the accelerating branch, so probabilities there fall but do not vanish.

The tabulated values are the fractions of posterior samples above each threshold. Fig. 10b of the main text plots the same quantity as a continuous curve over sea level. Those curves are smoothed for display only, by replacing each sample with a Gaussian kernel whose width is one quarter of Scott's rule for that ensemble, about 17 mm for the fast-WAIS case and about 6 mm for the slow-WAIS case. At every threshold and warming trajectory in Table S5, the smoothed curves differ from the unrounded sample fractions by at most 0.3 percentage points.

Table S5: **Probability (%) of exceeding global mean sea-level thresholds by 2100.** Thresholds are relative to the pre-industrial baseline (≈1850–1900). The $p(\mathrm{S1}) = 90\%$ column is the slow-WAIS case, which keeps 10% weight on the accelerating branch; the $p(\mathrm{S1}) = 10\%$ column is the fast-WAIS case. Values are computed from posterior samples under each warming trajectory.

SLR threshold (m, pre-ind.)	Warming trajectory	SPECTRUM p(s1)=90%	SPECTRUM p(s1)=10%
0.5	1.8 °C	93.2	99.2
	2.7 °C	99.5	100.0
	3.6 °C	100.0	100.0
0.75	1.8 °C	6.4	48.9
	2.7 °C	24.6	71.2
	3.6 °C	67.5	92.7
1.0	1.8 °C	2.4	21.8
	2.7 °C	3.2	28.7
	3.6 °C	4.9	39.4
1.5	1.8 °C	0.8	7.2
	2.7 °C	0.9	8.6
	3.6 °C	1.1	10.3
2.0	1.8 °C	0.4	3.3
	2.7 °C	0.4	3.7
	3.6 °C	0.5	4.5

3 Coastal Adaptation Costs and Flood Damages

The main text converts GMSL rise into annual global coastal adaptation costs and annual global flood damages without further adaptation (right-hand axes of Fig. 11a and the associated text).

3.1 Adaptation cost

The total capital cost of coastal adaptation is estimated at $C_{\mathrm{cap}} = \$4\,\mathrm{T/m}$ of GMSL rise. This is derived from Tiggeloven et al. (2020), who estimate that 3.4% of the global coastline (∼54,400 km of ∼1.6 million km total) warrants dike heightening under cost-benefit optimality, at a unit cost of \$7 M per km per meter of dike height (the median cost raising dikes in New Orleans post-Hurricane Katrina). This yields ∼\$400 B/m for dike raising alone along the optimal 3.4% of coastline. We apply a conservative 10× multiplier to account for the full societal adaptation burden—including seawalls, nature-based solutions, managed retreat, and infrastructure relocation—across a broader fraction of the global coastline.

Adaptation is dominated by capital investment deployed over the planning horizon, so we express it as an annual cost, in the same units as the flood damages below, using the capital recovery factor (CRF):

$$c_{\mathrm{a}} = C_{\mathrm{cap}} \times \mathrm{CRF}, \qquad \mathrm{CRF} = \frac{r\,(1+r)^n}{(1+r)^n - 1}, \tag{2}$$

where $r = 0.03$ is the real discount rate and $n = 75\,\mathrm{yr}$ is the planning horizon. This gives $\mathrm{CRF} \approx 0.034$ and an annualized adaptation cost of $c_{\mathrm{a}} \approx \$135\,\mathrm{B/(m{\cdot}yr)}$ of GMSL rise.

3.2 Flood damages

Flood damages are computed from the estimates of Jevrejeva et al. (2018), which relate GMSL rise to annual global flood costs without additional adaptation, based on estimated asset exposure. We interpolate linearly between six points. Four are values stated in their text, \$10.2, 11.7, 14.3, and 27.0 T/yr at 0.52, 0.63, 0.86, and 1.80 m of rise. One is zero cost at zero rise. The sixth, \$2.5 T/yr at 0.20 m, is read from their Fig. 4a, where the curves for their 1.5 °C, 2 °C, and RCP8.5 median scenarios pass through \$2.3–2.7 T/yr at that level of rise. Beyond 1.80 m we extrapolate linearly using the slope of the final segment, \$13.5 T/yr per meter.

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